

**RESOLVING MORPHOLOGICAL OVERLAP IN INFECTIOUS SKIN
DISEASES: AN ITA-GUIDED ADAPTIVE L*-CLAHE AND
FINE-GRAINED YOLOv26 FRAMEWORK FOR LOCALIZATION
AND CLASSIFICATION ON MELANIN-RICH FILIPINO SKIN**

A Thesis

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by

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Chapter 1

THE PROBLEM AND ITS SETTING

This chapter contains a broader view to the study's Introduction, Theoretical Framework, Conceptual Framework, Statement of the Problem, Hypothesis, Scope and Limitation, and Significance of the Study.

Introduction

Early detection of skin diseases is critical to patient outcomes, yet current AI-driven diagnostic tools exhibit significant performance disparities across demographic groups. Modern object detection architectures, including the latest generation represented by YOLOv26, have achieved remarkable improvements in computational efficiency and real-time localization accuracy. Despite recent advances in dermatological AI, accurate localization of infectious skin lesions on melanin-rich skin remains a challenging computer vision problem due to four interconnected computational challenges: illumination-induced feature degradation, morphological overlap between visually similar infectious diseases, skin tone/domain variation, and class imbalance. These challenges remain important factors that can reduce the reliability and generalizability of dermatological AI systems, particularly on underrepresented populations such as individuals with melanin-rich skin tones. The central question this study dares to answer is whether a unified computational pipeline can overcome all four of these challenges simultaneously, with particular emphasis on improving localization performance for melanin-rich skin within the Philippine clinical context.

Automated skin lesion detection and classification (traditionally dominated by melanoma detection research) has achieved impressive accuracy rates on predominantly light-skinned populations (Esteva et al., 2017, as cited in Joerg et al.,

2025). However, many early deep-learning skin lesion classification models exhibit substantial performance degradation when applied to darker or melanin-rich skin tones. This degradation is not incidental. It is the direct and measurable consequence of illumination-induced feature degradation, wherein melanin pigmentation reduces the luminance contrast between lesion boundaries and surrounding skin, making it structurally difficult for models trained on light-skin data to extract meaningful visual features from darker complexions (Benčević et al., 2024). Chamarthi et al. (2023) demonstrated that model sensitivity for melanoma detection plummets from 67% on HAM10000 (a predominantly light-skin dataset) to merely 11% when evaluated on darker skin tones, revealing critical fairness gaps in existing diagnostic systems. This disparity is not a limitation of deep learning itself, but rather a systematic failure rooted in dataset bias: existing large-scale dermatological datasets exhibit an imbalance ratio of 7.57, with Fitzpatrick skin types V–VI (the darkest skin tones) severely underrepresented (Chamarthi et al., 2023).

The second, and central, systemic challenge is morphological overlap between visually similar infectious diseases — a Fine-Grained Visual Categorization (FGVC) problem, a class of computer vision tasks in which subtle, low-level differences must be distinguished within a broader category of visually similar objects. Object detectors do not fail on melanin-rich skin simply because the image is dark; they fail because infectious lesions frequently share overlapping surface characteristics that become harder to separate once luminance contrast is degraded. This challenge is well documented in dermatological AI: Eskandari and Sharbatdar (2024), wherein it is found that psoriasis and lichen planus are so visually alike that automated systems routinely misclassify one for the other despite being in distinct conditions, while Sun et al. (2025) demonstrated that even subtle shape-based features, such as lesion boundary

sphericity, can be the deciding factor in separating one infectious skin condition from other visually similar lesions. This study deliberately selects eight infectious skin conditions that cluster into two overlapping morphological groups. The first is a vesiculopapular and eruptive exanthem cluster: Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, Molluscum Contagiosum, and Impetigo (*Impetigo Contagiosa (Mamaso)*), all of which present with fluid-filled or raised papular lesions that progress through visually similar developmental stages and are routinely flagged by Philippine health authorities as requiring active differential diagnosis from one another during outbreak surveillance, since early-stage lesions across these conditions are frequently indistinguishable to the eye. The second is a papulosquamous and verrucous cluster: Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*) which share scaling, hyperkeratotic texture, and patchy or circumscribed lesion geometry. Standard contrast enhancement techniques such as CLAHE compound this difficulty rather than resolving it, because they apply a single, fixed clip limit regardless of an image's underlying luminance distribution (George & Periyasamy, 2026; Otsuka et al., 2026). Because melanin-rich skin has a fundamentally lower baseline contrast than lighter skin, a static clip limit is mathematically miscalibrated across the input distribution: it under-enhances dark skin, leaving lesion crusts, scaling, and borders indistinguishable, or over-enhances light skin, amplifying sensor noise. When luminance contrast is further degraded by melanin pigmentation, the already-subtle distinguishing features among these eight diseases collapse further, increasing the likelihood that a detection model confuses one for another within its morphological cluster. Existing detectors therefore fail to distinguish visually similar infectious lesions not simply because the image is dark, but because their feature representations overlap when obscured by low-luminance skin.

The third systemic challenge is skin tone/domain variation — the phenomenon wherein a model trained on one population distribution degrades in performance when deployed on a meaningfully different one, particularly across differing skin tone distributions. Because many dermatological AI systems have been developed using datasets that do not adequately reflect the visual characteristics of melanin-rich skin, their generalization performance can degrade when applied to underrepresented populations. According to Daneshjou et al. (2022), algorithms trained exclusively on light skin perform significantly worse on Brown and Black skin, highlighting the urgent need for dataset diversification and architecture-level intervention. This challenge is particularly consequential for populations in the Philippines and across Southeast Asia, where melanin-rich skin is the norm and fungal and viral skin infections are prevalent (Chauhan et al., 2023). The burden falls particularly on dermatology practices in resource-limited settings, where specialist availability is scarce and diagnostic speed directly impacts patient prognosis.

The fourth systemic challenge is class imbalance. Even within a carefully curated eight-disease dataset, real-world prevalence and public data availability differ substantially across conditions — some of the target diseases, such as Warts, are extensively documented in existing dermatological repositories. When standard loss functions are applied during model training, the optimization process gravitates toward majority classes because they contribute disproportionately to gradient updates. Minority disease categories (despite their clinical significance) are effectively deprioritized by the mathematical structure of conventional training objectives, resulting in a model that performs well on aggregate metrics while failing precisely on the underrepresented cases that demand the most diagnostic attention (Lin et al., 2024).

This thesis addresses this convergence of challenges by developing the ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework for the localization and classification of eight morphologically overlapping infectious skin diseases: Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, Molluscum Contagiosum, Impetigo (*Impetigo Contagiosa (Mamaso)*), Ringworm (*Tinea Corporis (Buni)*), *An-an (Tinea Versicolor)*, Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*)— with targeted optimization for melanin-rich Filipino skin, evaluated across Fitzpatrick Skin Types I–V with primary analytical emphasis on Fitzpatrick Skin Types III–V. Rather than applying a fixed contrast-enhancement parameter to every image, the framework computes each image's Individual Typology Angle (ITA) from its CIELAB color values and dynamically scales the CLAHE clip limit as a function of that score, expressed as $clip_limit = f(ITA)$, through an ITA-Guided Adaptive L*-CLAHE module. This preprocessing step feeds into a sequential, temporally decoupled training strategy: Stage 1 trains the network on spatial localization (lesion boundaries, geometry, and position) after which selected backbone layers are frozen so that Stage 2 can concentrate on fine-grained morphological representation learning (texture, crusting, scaling, edges, and shape) without disrupting the localization features already learned. The framework further integrates Focal Loss to address class imbalance and Grad-CAM to validate that predictions are driven by genuine lesion morphology rather than background artifacts. By selecting eight diseases whose visual similarity constitutes two genuine fine-grained classification clusters, this study shifts its central contribution away from breadth of disease coverage and toward the depth of a mathematically grounded solution to morphological overlap under variable skin luminance.

Theoretical Framework

The ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework

The proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework serves as the theoretical foundation of this study. It is established upon the integration of complementary theories in luminance normalization, color-space transformation, skin-tone characterization, transfer learning, representation learning, fine-grained visual categorization, temporal training separation, imbalance-aware optimization, explainable artificial intelligence, and algorithmic fairness to address computational challenges in infectious skin lesion localization and classification. Contemporary dermatological artificial intelligence systems have demonstrated significant potential in automated lesion analysis. However, these performances may remain susceptible to illumination-induced feature degradation, domain shift across diverse skin tones, morphological similarity among infectious diseases, and long-tail class imbalance.

Rather than relying on a single computational technique, the proposed framework combines multiple theoretical foundations that collectively improve lesion visibility, feature specialization, disease discrimination, balanced learning, and model interpretability. CIELAB and adaptive L*-CLAHE provides the theoretical basis for enhancing luminance while preserving chromatic information, while ITA provides a quantitative measure of skin-lightness characteristics that guides the adaptive selection of the CLAHE clipping parameter. YOLOv26 transfer learning enables previously learned visual representations to be adapted to dermatological image analysis, while sequential training and layer freezing separate localization representation learning from classification-focused learning.

Fine-grained representation learning provides a basis for distinguishing diseases with overlapping morphological characteristics, while Focal Loss emphasizes difficult and underrepresented disease categories. Finally, Grad-CAM supports model interpretability, while algorithmic fairness and data representativeness provide a basis for evaluating model performance across diverse skin-tone groups. Collectively, these theoretical components establish a unified computational foundation for infectious skin lesion localization and classification across Fitzpatrick Skin Types I–V, with primary emphasis on melanin-rich skin (Fitzpatrick Skin Types III–V).

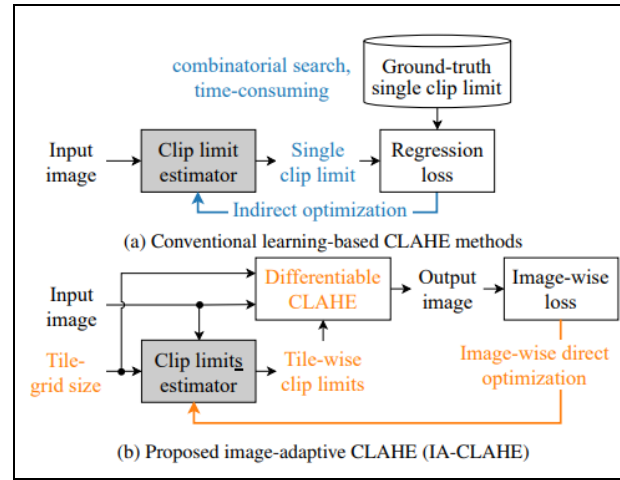


Figure 1. General theoretical illustration of the CIELAB, ITA, and Adaptive L*-CLAHE Framework (Adapted from Otsuka et al., 2026)

Luminance Normalization and Chromatic Preservation Through CIELAB, ITA, and Adaptive L*-CLAHE

The first theoretical foundation of the proposed framework is Luminance Normalization and Chromatic Preservation, which is grounded in color-space transformation and image enhancement principles. Dermatological image analysis is highly dependent on the visibility of lesion boundaries, surface texture, shape, and other structural characteristics. Variations in illumination and differences in skin pigmentation

may reduce the luminance contrast between lesions and surrounding skin, making important visual features more difficult for an object detection model to extract. Consequently, improving luminance representation while preserving relevant chromatic information provides a theoretical basis for producing more suitable image representations for dermatological analysis.

The framework adopts the CIELAB color space as the basis for separating luminance information from chromatic information. Unlike the RGB representation, CIELAB independently represents lightness through the L^* channel and chromatic information through the a^* and b^* channels. This separation allows luminance enhancement to be performed primarily on L^* while retaining the original chromatic components. The framework further incorporates the Individual Typology Angle (ITA) as a quantitative representation of skin-lightness characteristics. ITA is derived from the L^* and b^* components and provides an image-based measure that can be used to characterize differences in skin lightness.

To extend conventional L^* -CLAHE, the proposed framework introduces an ITA-guided adaptive enhancement mechanism. Instead of applying one fixed CLAHE clipping parameter to all images, the clipping parameter is represented as $clip_limit = f(ITA)$. The function is intended to be empirically calibrated using the study dataset rather than arbitrarily predetermined. This theoretical approach transforms L^* -CLAHE from a static preprocessing operation into an adaptive luminance-normalization mechanism that responds to differences in skin-lightness characteristics before lesion localization and classification.

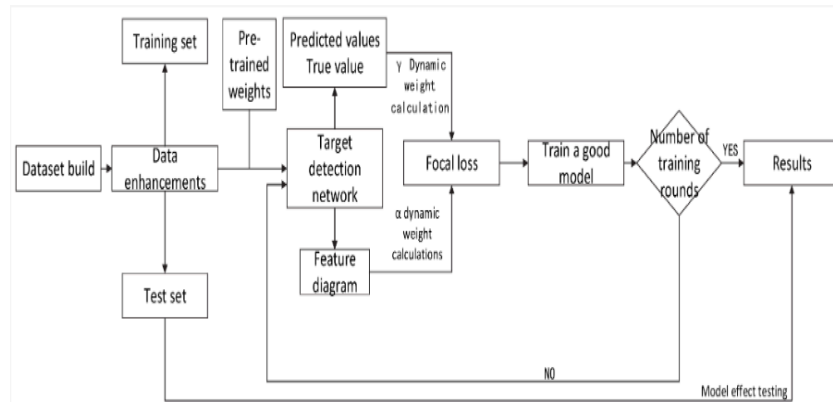


Figure 2. General theoretical illustration of YOLOv26 Transfer Learning (Adapted from Li et al., 2022)

Transfer Learning and YOLOv26-Based Decoupled Learning

The second theoretical foundation of the proposed framework is Transfer Learning and YOLOv26-Based Decoupled Learning. Transfer learning is founded on the principle that knowledge acquired from large-scale visual datasets can be transferred and adapted to specialized application domains. Rather than learning visual representations entirely from the beginning, a pretrained YOLOv26 model provides previously learned representations of general visual characteristics such as edges, textures, contours, shapes, and spatial structures. This theoretical principle supports efficient adaptation of a pretrained detection model to infectious skin lesion analysis.

Within the proposed framework, transfer learning is integrated with YOLOv26, which provides the primary architecture for lesion localization and disease classification. The framework adopts a sequential or temporally separated learning strategy because localization and classification have different objectives. Localization focuses on determining where the lesion is, whereas classification focuses on determining what disease the lesion represents. Separating these objectives allows the model to first

establish spatial representations before developing more specialized disease-related representations.

The first training stage therefore focuses on localization representation learning, where the model learns characteristics such as lesion position, boundaries, geometry, edges, general texture, and spatial structure. After the desired localization representations have been established, selected backbone layers are frozen before proceeding to the classification-focused stage. This theoretical combination of transfer learning, YOLOv26, and sequential training provides a basis for developing specialized representations while reducing potential optimization conflict between spatial localization and disease-specific classification objectives.

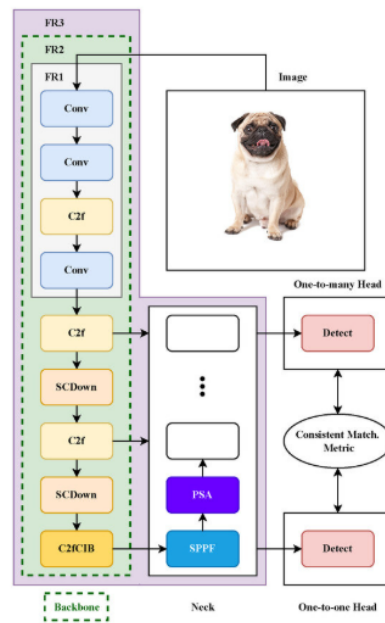


Figure 3. General theoretical illustration of Layer Freezing and Temporal Training Separation (Adapted from Dobrzycki et al., 2025)

Layer Freezing and Temporal Training Separation

The third theoretical foundation of the proposed framework is Layer Freezing and Temporal Training Separation, which is grounded in the principle of preserving useful learned representations while allowing subsequent layers to adapt to a specialized target task. In transfer learning, freezing selected layers can maintain previously acquired low-level and spatial representations while limiting unnecessary changes during later stages of training. Within the proposed framework, this principle is used as the transition mechanism between localization representation learning and classification-focused representation learning.

During Stage 1, the YOLOv26 model develops representations related to lesion localization, including spatial position, boundaries, edges, geometry, and overall lesion structure. Once this representation-learning stage reaches the intended training state, selected backbone layers are frozen. The purpose is to preserve the spatial information established during localization learning before the model is exposed to the more disease-specific requirements of the second stage.

This makes layer freezing an important architectural component rather than simply a conventional transfer-learning setting. By preserving localization-oriented representations while allowing subsequent learning to focus on disease-specific characteristics, temporal training separation provides a theoretical basis for improving feature specialization between lesion localization and infectious disease classification.

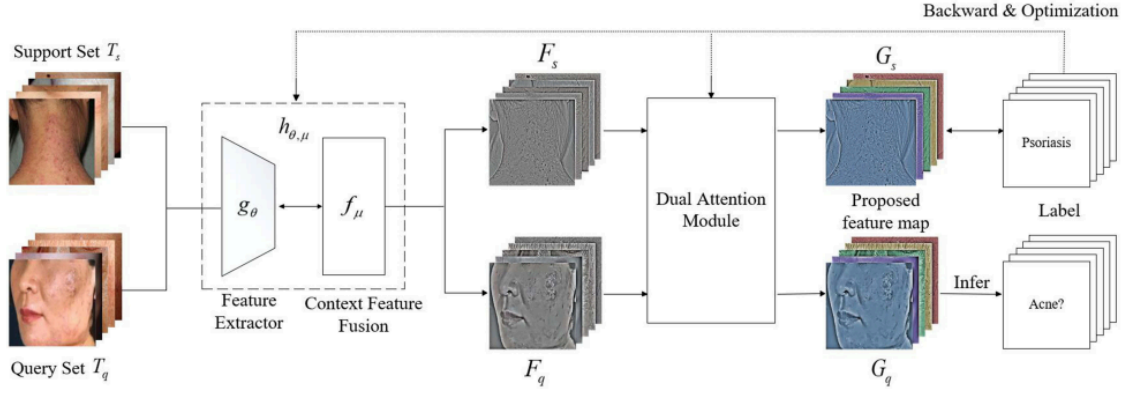


Figure 4. General theoretical illustration of Fine-Grained Morphological Representation Learning (Adapted from Chen et al., 2024)

Fine-Grained Representation Learning for Morphological Discrimination

The fourth theoretical foundation of the proposed framework is Fine-Grained Representation Learning, which addresses the challenge of morphological overlap among infectious skin diseases. Different infectious diseases may exhibit similar visual characteristics, including crusting, scaling, irregular or circular shapes, comparable textures, and overlapping lesion boundaries. As a result, generalized visual representations may not contain sufficient discriminative information to reliably distinguish diseases with similar appearances.

Following the localization-focused training stage and layer freezing, the second training stage focuses on learning fine-grained morphological representations. The model is intended to learn detailed characteristics such as texture, crust, scaling, edges, shape, lesion boundaries, surface irregularities, and spatial arrangement. These characteristics provide more specific information about the structure and appearance of infectious lesions and can assist in distinguishing diseases that share broad visual patterns.

The proposed approach does not introduce a separate handcrafted morphological classifier. Instead, the YOLOv26 representation-learning process is trained to become more sensitive to disease-specific morphological characteristics. This provides a theoretical basis for improving classification among visually similar infectious skin diseases by allowing the model to learn subtle differences that may not be captured by general feature representations alone.

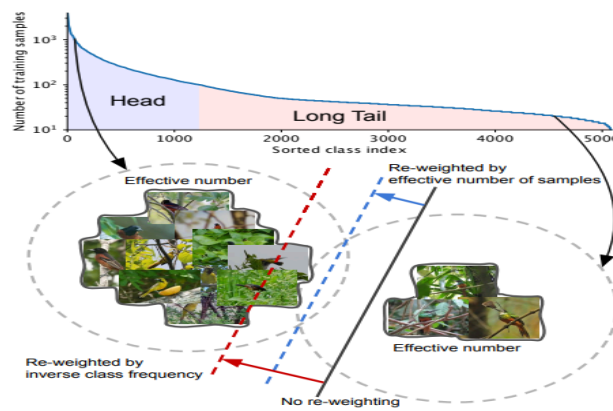


Figure 5. General theoretical illustration of long-tail class imbalance and Focal Loss Optimization (Adapted from Lin et al., 2024)

Focal Loss Optimization

The fifth theoretical foundation of the proposed framework is Focal Loss Optimization, which provides the theoretical basis for addressing long-tail class imbalance in dermatological datasets. Infectious skin disease datasets may contain unequal numbers of images across disease categories, resulting in some diseases being substantially underrepresented compared with majority categories. When conventional loss functions are applied, easily classified majority-class samples may contribute

substantially to the optimization process, potentially reducing the model's sensitivity to difficult and minority disease categories.

Focal Loss is grounded in the principle of imbalance-aware learning, wherein the contribution of easily classified examples is reduced while greater emphasis is placed on difficult examples. This theoretical approach encourages the learning process to allocate greater attention to minority disease categories, thereby improving class representation and reducing prediction bias. Within the proposed framework, Focal Loss complements the decoupled learning strategy by strengthening the theoretical foundation for balanced disease recognition while allowing lesion 10 localization and disease classification to remain conceptually independent learning objectives.

The application of Focal Loss also supports fairness in dermatological artificial intelligence by improving recognition performance across disease categories with varying prevalence. This theoretical perspective is consistent with the work of Lin et al. (2024), who introduced Focal Loss to address class imbalance in dense object detection, and with more recent dermatological studies emphasizing the importance of equitable artificial intelligence systems for underrepresented clinical presentations. Consequently, Focal Loss provides the theoretical basis for improving balanced learning and reducing performance disparities caused by long-tail disease distributions.

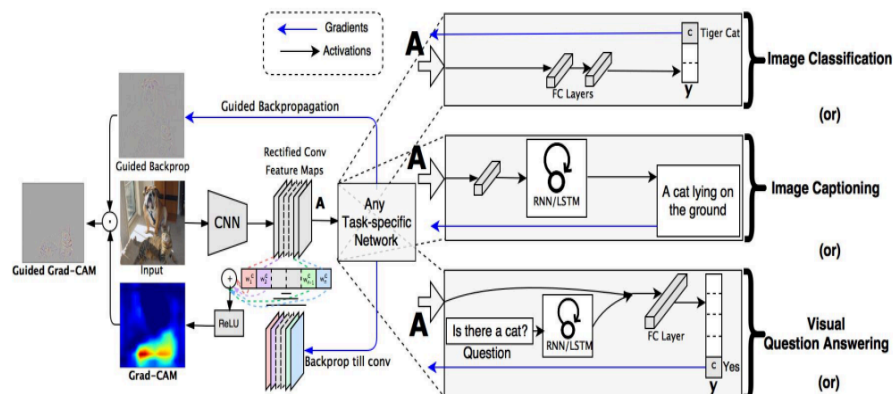


Figure 6. General theoretical illustration of Grad-CAM Visual Explanations

(Adapted from Selvaraju et al., 2017)

Explainable Artificial Intelligence (XAI)

The sixth theoretical foundation of the proposed framework is Explainable Artificial Intelligence (XAI). As artificial intelligence becomes increasingly applied to healthcare and dermatological analysis, the interpretability of model predictions becomes an important consideration. Deep learning models may generate accurate predictions without directly revealing which image characteristics influenced their decisions. Therefore, visual explanation techniques can provide additional evidence for determining whether the model is using meaningful lesion information.

The proposed framework adopts Gradient-weighted Class Activation Mapping (Grad-CAM) as its mechanism for visual explanation. Grad-CAM uses activation information from the model to generate class-specific heatmaps that identify image regions contributing to a prediction. These visualizations can be examined to determine whether the model's attention corresponds to relevant lesion structures such as boundaries, texture, crusting, scaling, shape, and other morphological characteristics.

The integration of Grad-CAM provides a theoretical basis for evaluating whether the representations learned during the fine-grained classification stage are associated with relevant lesion regions. The process can be represented as Prediction → Grad-CAM → Activation Heatmap → Lesion/Morphological Region Validation. Rather than functioning as a diagnostic algorithm itself, Grad-CAM provides visual evidence of model attention, supporting the evaluation of model behavior and helping identify

whether predictions are influenced by relevant lesion characteristics or potentially irrelevant regions such as background skin, shadows, and imaging artifacts.

Algorithmic Fairness and Data Representativeness

The seventh theoretical foundation of the proposed framework is Algorithmic Fairness and Data Representativeness, which establishes the theoretical relationship between dataset composition and the generalization capability of artificial intelligence models. Machine-learning systems learn representations from the statistical characteristics of their training data. When particular disease categories or skin-tone groups are insufficiently represented, the learned feature distributions may become biased toward majority categories, potentially reducing model performance for underrepresented groups.

The proposed framework includes Fitzpatrick Skin Types I–V, with primary emphasis on Types III–V, to evaluate model performance across different skin-tone groups. This consideration is important because differences in pigmentation can influence image luminance, lesion contrast, and the visual characteristics available for feature extraction. Adequate representation across skin types and disease categories therefore provides a theoretical basis for evaluating whether the learned representations generalize beyond a limited population or image distribution.

From this perspective, data representativeness supports the evaluation of domain-shift robustness and performance across Fitzpatrick Skin Types I–V. The framework does not treat overall accuracy as the only indicator of model effectiveness; instead, performance across skin-tone groups and disease categories is considered

when evaluating robustness. This theoretical foundation complements the ITA-guided adaptive enhancement strategy by providing a basis for determining whether the proposed system can maintain reliable lesion localization and classification across diverse skin characteristics, particularly under the challenging conditions associated with melanin-rich skin.

Integration of the Theoretical Foundations

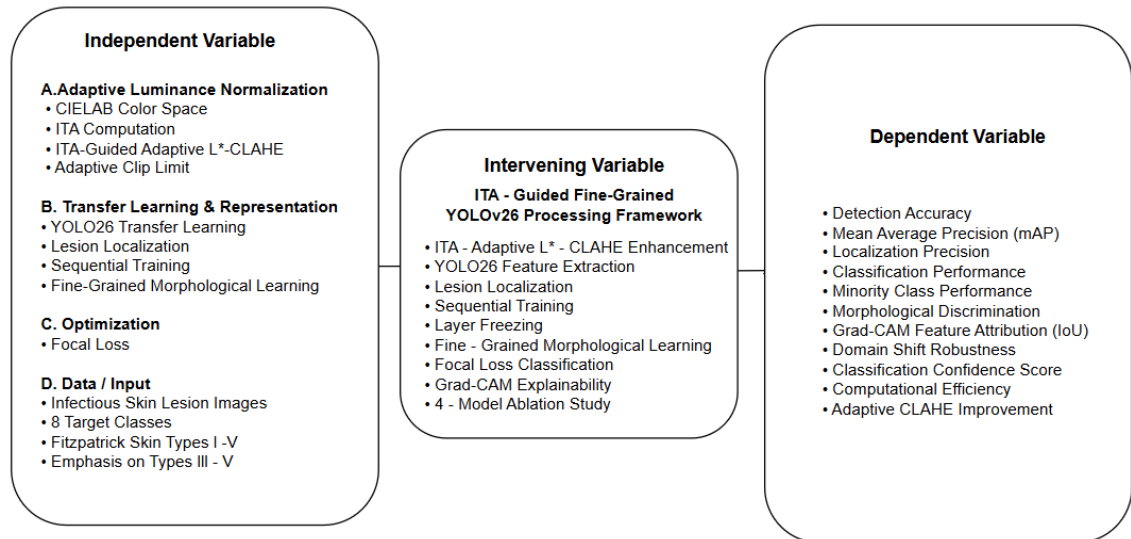
The proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework is founded on seven complementary theoretical principles: Luminance Normalization and Chromatic Preservation Through CIELAB, ITA, and Adaptive L*-CLAHE; Transfer Learning and YOLOv26-Based Decoupled Learning; Layer Freezing and Temporal Training Separation; Fine-Grained Representation Learning; Focal Loss Optimization; Explainable Artificial Intelligence; and Algorithmic Fairness and Data Representativeness. Together, these foundations address the major computational challenges identified in infectious skin lesion analysis.

The theoretical flow begins with CIELAB transformation and ITA estimation, where the ITA value is used to determine an adaptive CLAHE clipping parameter through the empirically calibrated relationship $clip_limit = f(ITA)$. The enhanced L* channel is recombined with the chromatic channels and reconstructed into RGB before being processed by YOLOv26. The first training stage develops localization representations, followed by selected backbone layer freezing and a second stage focused on fine-grained morphological representation learning. Focal Loss then supports balanced classification of difficult and underrepresented disease categories, while Grad-CAM provides visual interpretation of the model's predictions.

Rather than functioning as independent techniques, these theoretical foundations operate as interconnected components of a unified computational framework. Adaptive luminance normalization improves lesion visibility, transfer learning and temporal training separation support specialized feature learning, fine-grained representation learning addresses morphological overlap, Focal Loss mitigates long-tail class imbalance, Grad-CAM provides interpretability, and data representativeness supports evaluation across diverse skin-tone groups. Collectively, these foundations provide the theoretical basis for improving infectious skin lesion localization and classification across Fitzpatrick Skin Types I–V, with primary emphasis on melanin-rich skin (Fitzpatrick Skin Types III–V).

Conceptual Framework

Figure 7. Conceptual Framework



This study proposes an ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework for the localization and classification of infectious skin diseases across Fitzpatrick Skin Types I–V, with emphasis on Types III–V. The framework integrates CIELAB conversion, ITA estimation, ITA-guided adaptive L*-CLAHE, YOLOv26 transfer learning, sequential training, layer freezing, fine-grained morphological representation learning, Focal Loss, and Grad-CAM. The framework addresses illumination-related feature degradation, skin-tone domain shift, morphological similarity, and class imbalance. ITA is used to adapt the CLAHE clipping parameter through $clip_limit = f(ITA)$, while sequential training first establishes lesion localization representations before focusing on fine-grained disease-specific features. Selected backbone layers are subsequently frozen to preserve learned localization representations during classification-focused training. Focal Loss places greater emphasis on difficult and underrepresented classes, while Grad-CAM provides visual explanations of the model's predictions. The framework focuses on eight target infectious skin disease classes and is further evaluated through a 4-Model Ablation Study to examine the contribution of the major framework components to localization, classification, domain robustness, explainability, and computational performance.

Independent Variables

These variables represent the computational techniques, configurations, and dataset characteristics established and examined throughout the study.

A. Adaptive Luminance Normalization Variables

CIELAB Color Space Conversion. Converts RGB skin lesion images into the CIELAB color space, separating luminance information represented by the L* channel from

chromatic information represented by the a^* and b^* channels. This allows luminance enhancement to be performed primarily through the L^* channel while preserving the original chromatic characteristics of the lesion.

Individual Typology Angle (ITA). Calculates the Individual Typology Angle (ITA) from the CIELAB representation to characterize the lightness characteristics of the skin region. The resulting ITA value serves as the basis for determining the appropriate adaptive contrast-enhancement configuration.

ITA-Guided Adaptive L^* -CLAHE. Uses the calculated ITA value to determine the appropriate CLAHE clipping parameter rather than applying a single fixed clip limit to every image. The relationship is represented as $clip_limit = f(ITA)$. The mapping function will be empirically calibrated using the study dataset to establish an appropriate relationship between ITA and the CLAHE clipping parameter.

Adaptive Clip Limit. Determines the CLAHE clipping parameter according to the estimated ITA rather than applying a fixed clip limit across all input images. The selected clip limit changes according to the ITA characteristics of the image, allowing the contrast-enhancement strength to adapt to variations in skin luminance. The ITA-to-clip-limit mapping will be empirically calibrated using the study dataset before application within the proposed framework.

B. Transfer Learning and Representation Learning Variables

These variables describe how the YOLOv26 model is adapted and trained to first learn lesion localization and subsequently learn fine-grained morphological characteristics for disease classification.

YOLOv26 Transfer Learning. Uses a pretrained YOLOv26 architecture to provide hierarchical visual representations and adapt the model to the localization and classification of infectious skin lesions.

Localization Representation Learning. The first training stage focuses on learning spatial representations associated with lesion location, lesion boundaries, lesion geometry, edges, and spatial structure. This stage establishes representations necessary for accurately identifying and localizing the lesion within the image.

Sequential Training Separation. Separates the localization-focused learning stage from the subsequent classification-focused learning stage. The sequential approach first establishes lesion localization representations before emphasizing disease-specific morphological characteristics. This separation is intended to reduce optimization conflict between learning where the lesion is located and learning which disease the lesion represents.

Layer Freezing Strategy. After localization representation learning, selected backbone layers are frozen to preserve the learned spatial representations. The remaining trainable components can then adapt toward disease-specific classification without substantially altering the established localization features.

Fine-Grained Morphological Representation Learning. The second training stage focuses on learning detailed morphological characteristics that contribute to disease discrimination, including texture, crusting, scaling, edges, shape, lesion boundaries, and surface patterns. These representations are intended to improve discrimination among infectious skin diseases with visually similar morphological characteristics.

C. Optimization Variable

Focal Loss Optimization. Uses Focal Loss during classification-focused training to address long-tail class imbalance. The loss function places greater emphasis on difficult and misclassified samples while reducing the contribution of easily classified majority-class samples. This allows the model to provide greater learning attention to underrepresented infectious disease categories.

D. Data / Input Variables

These variables define the image data and target categories used as inputs to the proposed framework and establish the skin-type coverage of the study.

Infectious Skin Lesion Images. Uses qualified dermatological images of infectious skin diseases as the primary input of the proposed framework. The images provide the visual information required for lesion localization, fine-grained morphological representation learning, and disease classification.

Eight Target Classes. The study focuses on eight selected infectious skin disease classes as the target categories for lesion localization and classification. The same eight target classes are maintained across the experimental models, training procedures, ablation configurations, and evaluation protocols to ensure consistent comparison.

Fitzpatrick Skin Types I–V. Includes images representing Fitzpatrick Skin Types I–V to evaluate the ability of the proposed framework to perform across different skin-tone conditions. The inclusion of these skin types allows the study to examine the robustness of the framework under varying levels of skin pigmentation.

Emphasis on Fitzpatrick Skin Types III–V. Places particular analytical emphasis on Fitzpatrick Skin Types III–V, which represent the primary skin-tone conditions of interest in evaluating the framework's robustness under greater melanin representation. Performance across these skin types is examined alongside the broader Fitzpatrick I–V range to assess potential skin-tone domain differences.

Intervening Variable

ITA-Guided Fine-Grained YOLOv26 Processing Framework. The processing framework integrates ITA-guided adaptive L*-CLAHE, YOLOv26 feature extraction, lesion localization, sequential training, layer freezing, fine-grained morphological learning, Focal Loss classification, Grad-CAM explainability, and a 4-Model Ablation Study into a unified computational framework for infectious skin lesion localization and classification. These components work together to enhance lesion visibility, establish localization and disease-specific feature representations, address morphological similarity and class imbalance, preserve learned localization features, and provide interpretable model predictions. The 4-Model Ablation Study further evaluates the integrated effect of the major framework components by comparing model configurations under consistent experimental conditions.

Dependent Variables / Success Metrics

These variables represent the measurable outcomes used to evaluate the effectiveness of the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework across the eight target infectious skin disease classes and Fitzpatrick Skin Types I–V.

A. Detection and Localization Performance

- Detection Accuracy
- Mean Average Precision (mAP)
- Localization Precision

These measures evaluate the model's ability to accurately detect infectious skin lesions and identify their spatial locations within the input images.

B. Classification Performance

- Classification Performance
- Minority-Class Performance
- Classification Confidence Score

These measures evaluate the model's ability to correctly distinguish the eight target disease classes, particularly difficult and underrepresented categories.

C. Fine-Grained Morphological Discrimination

- Morphological Discrimination
- Morphologically Similar Disease Performance
- Misclassification Rate

These measures evaluate the model's ability to distinguish infectious skin diseases with visually similar morphological characteristics, including differences in texture, shape, boundaries, scaling, crusting, and surface patterns.

D. Domain Robustness

- Domain Shift Robustness
- Performance across Fitzpatrick Skin Types I–V

- Performance across Fitzpatrick Skin Types III–V

These measures evaluate whether the framework maintains consistent performance across different skin-tone conditions, with particular emphasis on Fitzpatrick Skin Types III–V.

E. Explainability

- Grad-CAM Feature Attribution (IoU)

This measure evaluates whether the regions highlighted by Grad-CAM correspond to relevant lesion areas and provide meaningful visual explanations of the model's predictions.

F. Computational Efficiency

- Computational Efficiency
- Inference Latency
- Frames Per Second (FPS)
- GFLOPs
- Memory Consumption

G. Adaptive CLAHE Improvement

- Adaptive CLAHE Improvement

This measure evaluates the performance improvement associated with the ITA-guided adaptive L*-CLAHE compared with the corresponding non-adaptive or baseline configuration within the 4-Model Ablation Study. This allows the contribution of

adaptive luminance normalization to be assessed based on measurable changes in model performance.

Statement of the Problem

This study aims to evaluate and compare the localization and classification performance of the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework (Model D) against baseline and ablation detection configurations (Models A–C) across diverse skin tones and target disease classes.

Specifically, it seeks to answer the following questions:

1. What is the baseline localization (mAP₅₀, mAP_{50–95}) and classification (Precision, Recall, F1-score) performance of the following model configurations across the 8 infectious skin disease classes: Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, Molluscum Contagiosum, Impetigo (*Impetigo Contagiosa (Mamaso)*), Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*) and stratified Fitzpatrick Skin Types?
 - Model A: Baseline YOLOv26
 - Model B (using a standard clip limit of 2.0): Fixed-Parameter L*-CLAHE YOLOv26
 - Model C: YOLOv26 with Focal Loss Optimization
 - Model D: ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework (Proposed Decoupled Two-Stage)
2. Which statistically significant differences exist in localization (mAP₅₀, mAP_{50–95}) and classification (Precision, Recall, F1-score) performance among Models A, B, C, and D across the 8 target infectious skin disease classes?

3. What performance improvement (ΔAP_{50}) is achieved by the proposed framework (Model D) when comparing light skin types (Fitzpatrick Skin Types I–II) versus melanin-rich skin types (Fitzpatrick Skin Types III–V) across all disease classes?

Hypothesis

The following null (H_0) and alternative (H_1) hypotheses are formulated and will be tested at a 0.05 level of significance:

Hypothesis 1 (Inter-Model Comparison).

Null Hypothesis (H_0)

There is no statistically significant difference in the localization (mAP₅₀, mAP_{50–95}) and classification (Precision, Recall, F1-score) performance among Models A, B, C, and D across the 8 infectious skin disease classes.

Alternative Hypothesis (H_1)

There is a statistically significant difference in the localization (mAP₅₀, mAP_{50–95}) and classification (Precision, Recall, F1-score) performance among Models A, B, C, and D across the 8 infectious skin disease classes.

Hypothesis 2 (Skin Tone Disparity Comparison for Proposed Model D).

Null Hypothesis (H_0)

There is no statistically significant difference in the performance improvement (ΔAP_{50} between baseline Model A and Proposed Model D) when evaluated between

lighter skin types (Fitzpatrick Skin Types I–II) and melanin-rich skin types (Fitzpatrick Skin Types III–V) across the 8 skin disease classes.

Alternative Hypothesis (H_1)

There is a statistically significant difference in the performance improvement (ΔAP_{50} between baseline Model A and Proposed Model D) when evaluated between lighter skin types (Fitzpatrick Skin Types I–II) and melanin-rich skin types (Fitzpatrick Skin Types III–V) across the 8 skin disease classes.

Scope of the Study

This study develops and evaluates the ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework for the localization and classification of eight morphologically overlapping infectious skin diseases on melanin-rich Filipino skin. The research is primarily limited to investigating how ITA-guided adaptive luminance normalization, decoupled transfer learning, class-imbalance optimization, and explainable artificial intelligence techniques can resolve fine-grained morphological overlap and improve lesion localization accuracy, classification performance, and model interpretability within a controlled computational environment.

- **Geographic and Demographic Scope.** Investigation is focused on the Philippine clinical context, with particular attention to skin disease presentations commonly encountered among Filipino patients. The study covers the full Fitzpatrick Skin Types I–V (via image-based proxy) spectrum to enable comparative evaluation across skin tones, with primary analytical focus remaining on Fitzpatrick Skin Types III–V (the skin types most prevalent among Filipinos), as the proposed pipeline is specifically designed to address the performance

disparities experienced by this demographic group. Furthermore, Fitzpatrick Type VI is deliberately excluded as it falls outside the typical skin-tone range of the target Filipino demographic.

- **Disease Scope.** The study addresses eight infectious skin disease subtypes, selected because they cluster into two groups exhibiting severe morphological overlap (framed analytically as a testable study premise rather than an absolute clinical fact):
 - Vesiculopapular / Eruptive Exanthem Cluster (4): Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, Molluscum Contagiosum, and Impetigo (*Impetigo Contagiosa (Mamaso)*).
 - Papulosquamous / Verrucous Cluster (4): Ringworm (*Tinea Corporis (Buni)*), *An-an (Tinea Versicolor)*, Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*).

This selection is a deliberate methodological choice: rather than training a detector to distinguish diseases that are already visually dissimilar, the study concentrates its computing problem on two clusters of diseases the model is most likely to confuse within each cluster, directly testing whether the proposed Adaptive ITA-CLAHE algorithm can expose distinguishing morphological detail that a static preprocessing baseline cannot. These clusters are utilized strictly as analytical groupings for fine-grained computer vision evaluation, not as official medical or diagnostic classifications.

- **Data Privacy Scope.** The study strictly relies on anonymized data. No direct personal identifiers (names, addresses) will be collected, though redacted sensitive health information (diagnoses and certificates) will be securely processed. The only inputs retained from clinical hold-out respondents are the

submitted skin image, its corresponding disease label, and a fully redacted medical certificate used strictly for diagnostic verification.

- **Anatomical and Technical Scope.** Lesions are restricted to accessible body sites excluding the face. Images must meet quality standards: minimum 2000 × 2000 pixels, perpendicular angle to skin surface.
- **Dataset Composition Scope.** The global foundation total target dataset is 300–500 images per disease subtype, yielding 2,400–4,000 total images across the eight subtypes. The minimum threshold is 300 total verified images per disease type across the global dataset prior to the 70/20/10 split to ensure sufficient data for a meaningful detection task. The dataset is composed of two strategically combined sources. The global foundation (comprising approximately 90–95% of the total dataset) is sourced from publicly available dermatological repositories, including DermNet and Kaggle, providing representation across Fitzpatrick Skin Types I–V (via image-based proxy). The clinical hold-out set comprises approximately 5–10% of the total dataset (an estimated 120 to 400 images, distributed across the eight disease subtypes, approximately 15 to 50 images per disease), consisting of locally sourced Philippine clinical images collected from consenting Filipino respondents via a structured Google Form instrument, supplemented by global repository data where necessary. This hybrid composition ensures both sufficient data volume for model training and a real-world domain shift stress test using the genuinely local Philippine subset. The global foundation (public repository) portion of the dataset follows a 70/20/10 stratified split into training, validation, and public testing subsets across all disease types. The clinical hold-out set is kept entirely separate from this split and is never used during training or validation; it is reserved exclusively as an external test set to evaluate real-world domain shift.

- Task Scope.** The study focuses on object detection (lesion localization and classification) not classification alone, with particular emphasis on resolving classification confusion within each of the two morphological clusters. Detection models must identify lesion bounding boxes and assign disease categories. With that, Human-computer interaction (HCI) evaluations, usability testing, and clinical safety audits of the Patient-Facing View are beyond the scope of this computer vision study and are reserved for future work. Localization and classification are further separated into a sequential Stage 1 (spatial localization) and Stage 2 (fine-grained morphological classification) training procedure. Model development is limited to detection accuracy, fine-grained classification accuracy among the eight target diseases, and computational efficiency metrics suitable for mobile and edge deployment. Explainability analysis is restricted to Grad-CAM visualization.
- Computational and Methodological Scope.** The study employs a hybrid computational evaluation design, wherein model performance is evaluated on both held-out test data drawn from publicly available datasets and a purposively sampled clinical hold-out set of locally sourced Philippine images, constituting a real-world domain shift stress test. Transfer learning is restricted to YOLOv26 backbone initialized with COCO dataset weights. Data augmentation is limited to standard geometric augmentation and ITA-Guided Adaptive L*-CLAHE preprocessing, wherein the CLAHE clip limit is computed as a function of each image's Individual Typology Angle (ITA), applied ahead of the Stage 1/Stage 2 training procedure. Focal Loss hyperparameters will be initialized using $\alpha = 0.25$ and $\gamma = 2.0$ as reference starting values. Their optimal configuration will subsequently be determined through grid-search validation using the training and validation data. Furthermore, the experimental evaluation is strictly delimited to a

four-model ablation study— comprising the Baseline YOLOv26 (Model A), YOLOv26 with fixed L*-CLAHE (Model B), YOLOv26 with Focal Loss (Model C), and the fully integrated proposed framework (Model D)— to quantitatively evaluate the integrated and combined effects of the framework components.

Limitations of the Study

To maintain methodological consistency, dataset quality, and research focus, the study acknowledges the following limitations:

- Dataset Availability and Representation.** The global foundation of the dataset relies on publicly available dermatological repositories (specifically DermNet and Kaggle) supplemented by a locally sourced clinical hold-out set. Preliminary source-availability scouting indicates strong dataset support for four of the eight target diseases; Warts (*Verruca(Kulugo)*), Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), and Hand, Foot, and Mouth Disease in particular have substantial existing coverage, while the other four, Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Athlete's Foot (*Tinea Pedis (Alipunga)*), and Impetigo (*Impetigo Contagiosa (Mamaso)*) have partial existing coverage that falls short of or close to the minimum threshold and will require supplementary source searching beyond the primary repositories to reach the target range. If adequate images cannot be sourced for a particular disease despite exhaustive searching, that subtype will be excluded and, where feasible, substituted with a disease of comparable morphological overlap within the same cluster.
- Restricted Demographic Scope.** Although the study includes Fitzpatrick Skin Types I–V to enable cross-skin-tone comparative evaluation, the primary

analytical focus and pipeline optimization remain centered on Fitzpatrick Skin Types III–V. Consequently, the findings and performance improvements reported by this study are most directly applicable to melanin-rich skin populations and may require additional validation before generalization to broader Fitzpatrick I–II populations or populations outside the Philippine clinical context.

- Disease Coverage Skewed Toward Viral Etiology.** The final eight-disease set spans three pathogenic categories: (1) *four viral*: Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, and Warts (*Verruca(Kulugo)*), (2) *three fungal*: Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Athlete’s Foot (*Tinea Pedis (Alipunga)*), and (3) *one bacterial*: Impetigo (*Impetigo Contagiosa (Mamaso)*). No parasitic conditions are represented. This composition was driven primarily by the strength of morphological overlap within each cluster and by practical data availability rather than an intent to represent the full pathogenic diversity of Philippine dermatological presentations, and performance on diseases outside the two identified morphological clusters remains beyond the scope of this research.
- Single-Architecture Evaluation.** The study evaluates only the YOLOv26 architecture combined with ITA-Guided Adaptive L*-CLAHE preprocessing, decoupled Stage 1/Stage 2 training, Focal Loss optimization, and Grad-CAM visualization. Alternative object detection architectures are not investigated.
- Clinical Hold-Out Set Constraints.** The locally sourced clinical hold-out set, targeted at approximately 5–10% of the total dataset (120–400 images across eight disease subtypes), relies on fully redacted medical certificates for diagnostic ground truth. While dermatologist validation is applied to the hold-out set, it remains a constrained sample of Philippine clinical data and functions as a

qualitative domain shift stress test rather than a comprehensive clinical validation.

- Algorithmic Calibration Boundary.** The mathematical function used to scale the contrast enhancement, defined as the $clip_limit=f(ITA)$, is empirically calibrated strictly for the dermatological images and varying skin tones within the study's specific dataset. Applying this exact mathematical threshold to non-dermatological images (such as general photography) or images captured under radically different hardware sensors may require recalibration. This establishes the boundary of the algorithm's intended clinical domain.
- Data Collection Risk Mitigation.** The study recognizes the risk that respondents may decline to submit sensitive medical images through the primary Google Form collection instrument, given the personal nature of the data requested. Should this channel fail to yield a sufficient volume of qualifying clinical images across the eight target diseases, particularly for Fitzpatrick Skin Types III–V, the study will supplement the shortfall with images drawn from open-access global repositories such as the ISIC archives. These supplementary images, however, are not genuinely local Philippine clinical photographs and do not carry the same real-world domain-shift evidentiary value as respondent-submitted data. Should ISIC or other global-repository images constitute a substantial portion of the final hold-out set, the study will transparently report the proportion of Filipino-sourced versus supplementary images, and any claims regarding Philippine-specific domain-shift robustness will be scoped only to the genuinely locally sourced subset of the hold-out set.
- Generalizability Constraints.** The findings are primarily applicable to the eight selected infectious skin diseases and the populations represented within the combined dataset. Additional validation using larger, more diverse datasets

(including bacterial and parasitic conditions) is necessary before broader application.

- **Clinical Application Constraints.** The proposed framework serves as an AI-assisted screening and localization tool designed to support, not replace, dermatological expertise. It does not provide diagnoses, prescribe treatments, or function as a standalone clinical decision-making system. All predictions must be verified by qualified dermatologists before clinical application.

Significance of the Study

This study is significant because it investigates the development and evaluation of an ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Computational Framework for the localization and classification of infectious skin lesions across Fitzpatrick Skin Types I–V, with particular emphasis on melanin-rich skin types III–V. The study focuses on determining whether image-adaptive luminance enhancement and fine-grained decoupled training can improve the performance of YOLOv26 under varying skin-tone conditions.

The study may benefit the following:

1. **Computer Science and Artificial Intelligence Researchers.** The study contributes to computer vision and artificial intelligence research by providing an experimental evaluation of an adaptive image-preprocessing and object-detection framework. The comparison among the four ablation configurations (Baseline YOLOv26, YOLOv26 with Fixed L*-CLAHE, YOLOv26 with Focal Loss, and the ITA-Guided Adaptive L*-CLAHE proposed framework) provides a basis for determining the integrated and combined effects of adaptive luminance processing and class-imbalance optimization to object detection performance.

2. **Computer Vision and Machine Learning Researchers.** The study may provide a reference for researchers investigating object detection under challenging visual conditions. In particular, the integration of ITA-guided luminance enhancement with YOLOv26 and fine-grained decoupled training may serve as a basis for future research involving illumination variation, morphological similarity, and skin-tone-related performance differences.
3. **Healthcare Professionals and Dermatologists.** The proposed framework may serve as an AI-assisted screening and lesion-localization tool by assisting in the localization and classification of infectious skin lesions from images. The results may provide insight into the potential of computer vision techniques to support dermatological image analysis. However, the system is intended to support rather than replace professional clinical judgment.
4. **Filipino and Melanin-Rich Populations.** The study places particular emphasis on Fitzpatrick Skin Types III–V. Evaluating model performance across skin-tone groups may provide evidence regarding whether the proposed computational framework maintains reliable localization and classification performance for melanin-rich skin. This may contribute to future development of more representative dermatological AI systems.
5. **Academic Institutions.** The study may serve as a reference for future undergraduate and graduate research involving artificial intelligence, computer vision, object detection, and medical image analysis.
6. **Future Researchers.** The proposed framework and evaluation methodology may provide a foundation for future studies involving larger and more diverse datasets, additional skin diseases, alternative object detection architectures, and other adaptive image-processing or fine-grained learning techniques.

Definition of Terms

- **Bounding Box Localization.** The spatial coordinates and boundaries predicted by the object detection model to specifically isolate and frame a skin lesion within an image.
- **CIELAB Color Space.** A color coordinate system used to mathematically separate an image's luminance (brightness) from its chromatic (color) data. This allows the system to enhance the visibility of lesions without altering their biologically relevant colors.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization).** An image preprocessing algorithm applied exclusively to the isolated L* channel to dynamically enhance local contrast, revealing subtle lesion textures and structural boundaries that are typically obscured by melanin-rich pigmentation or uneven lighting.
- **Class Imbalance.** A structural dataset problem where certain skin diseases (majority classes) are highly represented in training data, while others (minority classes) lack sufficient samples, causing the deep learning model to become biased against rarer conditions.
- **Domain Shift.** The degradation in a machine learning model's predictive performance when the real-world data it processes (e.g., varying skin tones, smartphone camera lighting) differs significantly from the controlled distribution of the data it was originally trained on.
- **Explainable Artificial Intelligence (XAI).** A validation mechanism integrated into the model to provide visual, objective evidence of its decision-making

process, ensuring the network focuses on clinically relevant lesion features rather than confounding background artifacts.

- **Fine-Grained Visual Categorization (FGVC).** A class of computer vision tasks in which subtle, low-level differences must be distinguished within a broader category of visually similar objects, such as morphologically overlapping infectious skin diseases.
- **Fitzpatrick Skin Types.** A standardized numerical classification schema for human skin color. The study targets skin tones with Fitzpatrick Skin Types I–V.
- **Focal Loss.** A modified gradient optimization function integrated into the detection head that dynamically reduces the learning weight of easily classified majority samples, thereby forcing the network to heavily penalize errors on difficult, underrepresented minority disease classes.
- **Grad-CAM (Gradient-weighted Class Activation Mapping).** An explainability technique that extracts activation gradients from the network's final convolutional layers to generate spatial heatmaps, mapping exactly which pixels influenced the model's prediction.
- **Individual Typology Angle (ITA).** A quantitative representation of skin-lightness characteristics derived from the L^* and b^* components of the CIELAB color space, used to dynamically determine the appropriate CLAHE clipping parameter for each image.
- **Intersection-over-Union (IoU).** A spatial evaluation metric used to mathematically quantify the alignment and overlap ratio between the model's generated Grad-CAM heatmaps and the actual ground-truth boundaries of the skin lesion.

- **L* Channel (Lightness Channel).** The specific isolated component within the CIELAB color space that dictates pixel brightness independently of the a* and b* color channels.
- **AP50 (Average Precision at IoU = 0.50).** A standard metric used to evaluate object detection models. A predicted bounding box is counted as a correct match (True Positive) if its overlap with the ground-truth box, measured by Intersection over Union (IoU), is at least 50%.
- **Mean Average Precision (mAP₅₀ and mAP₅₀₋₉₅).** The primary quantitative performance metric used to evaluate the exactness and precision of the model's bounding box predictions against the true location of the skin lesion.
- **Object Detection.** A dual-stage computer vision task that regression-maps image pixels directly to bounding box coordinates (localization) while simultaneously calculating class probabilities (classification) in a single forward pass.
- **Transfer Learning.** A computational strategy where a model utilizes generalized feature extraction capabilities (like edges and shapes) pre-trained on massive global datasets, which are then frozen and fine-tuned on smaller, localized dermatological datasets.
- **YOLOv26 (You Only Look Once, Version 26).** An advanced, end-to-end, NMS-free real-time object detection architecture that eliminates post-processing latency and serves as the baseline algorithmic engine for detecting and categorizing skin lesions in this framework.

Chapter 2

THE REVIEW OF RELATED LITERATURE AND STUDIES

This study cited books, articles, and journals that are pertinent to the recent study. It also provides explanations and logical connections between the present and past studies.

The Evolution of YOLO Architectures and Object Detection Frameworks.

Comparison of One-Stage and Two-Stage Object Detection Models

Sapkota et al. (2023) evaluated the performance of YOLOv8 and Mask R-CNN in complex orchard environments containing varying illumination conditions, object occlusions, and background clutter. The study aimed to examine how one-stage and two-stage object detection architectures differ in terms of detection capability and computational efficiency. Both models were trained and evaluated using the same dataset and performance metrics, including precision, recall, mean Average Precision (mAP), and inference speed.

The findings revealed notable differences between the two architectures. YOLOv8 demonstrated faster inference and stronger real-time performance, while Mask R-CNN provided more detailed object segmentation and region-level representation. These results highlight the trade-off between processing efficiency and segmentation capability, suggesting that architecture selection should be guided by application requirements rather than a single performance metric.

Despite providing a balanced comparison between one-stage and two-stage detectors, the study was conducted using agricultural imagery rather than medical datasets. The evaluation also focused primarily on detection and segmentation performance without examining challenges associated with medical imaging, such as

subtle lesion boundaries, demographic variability, and class imbalance. Consequently, further research is needed to determine whether similar architectural trade-offs persist in dermatological image analysis.

Evolution and Efficiency of YOLO-Based Detection Architectures

Ali et al. (2024) conducted a comprehensive review of object detection architectures, focusing on the evolution of the YOLO family and its comparison with competing frameworks such as Faster R-CNN, SSD, RetinaNet, and EfficientDet. The review synthesized findings from multiple application domains, including surveillance, industrial automation, autonomous systems, and healthcare, to identify major trends in object detection development.

The review found that different object detection architectures exhibit distinct strengths and limitations. YOLO-based models generally emphasized computational efficiency and real-time processing, whereas architectures such as Faster R-CNN often prioritized localization precision. The study also highlighted that architectural improvements introduced in later YOLO versions contributed to enhanced performance, although the effectiveness of these improvements varied depending on the dataset and application domain.

As a review-based investigation, the study relied on previously published findings rather than conducting direct experimental validation. Furthermore, it provided limited discussion regarding how architectural performance is affected by domain-specific challenges such as illumination variability, skin-tone diversity, and highly imbalanced datasets. These limitations indicate the need for empirical studies evaluating object detection architectures within specialized healthcare environments.

Performance Evaluation of Modern Object Detection Frameworks

Sharma et al. (2024) compared several contemporary object detection architectures, including YOLOv8, YOLOv9, YOLOv10, and Detectron2, using standardized datasets and evaluation procedures. The study sought to examine how recent object detection models differ in terms of detection accuracy, computational efficiency, and inference performance. To ensure consistency in evaluation, the researchers employed metrics such as precision, recall, F1-score, mean Average Precision (mAP), and processing time.

The findings of Sharma et al. (2024) showed that performance varied depending on the evaluation metric being considered. While YOLO-based models generally achieved faster inference speeds, other frameworks demonstrated competitive localization capabilities under specific conditions. Mahmood et al. (2024) further observed that improvements introduced in newer YOLO versions resulted in incremental gains rather than universally superior performance, emphasizing the continued presence of trade-offs among modern detection architectures.

Although Sharma et al. (2024) provided an updated comparison of contemporary object detection frameworks, the study was limited to benchmark datasets and did not investigate specialized medical imaging tasks. Furthermore, factors such as explainability, robustness to image-quality variation, and fairness across diverse demographic groups were not examined. These limitations suggest opportunities for future research involving clinical and dermatological image analysis.

Detection Accuracy and Computational Efficiency of Object Detection Models

Khalifa et al. (2025) evaluated YOLOv8, SSD, and Faster R-CNN to determine how different object detection architectures balance detection accuracy and computational efficiency. Using a common dataset and standardized evaluation metrics, the researchers conducted a direct comparison of the three models under identical experimental conditions. The investigation focused on metrics such as precision, recall, mean Average Precision (mAP), and inference speed.

According to Khalifa et al. (2025), each architecture demonstrated distinct strengths and weaknesses. Faster R-CNN achieved strong localization performance but required longer processing times due to its two-stage detection mechanism. SSD offered faster detection speeds but generally produced lower accuracy, whereas YOLOv8 provided a more balanced combination of efficiency and detection capability. These findings highlight the persistent trade-off between processing speed and predictive precision in object detection systems.

Despite the valuable insights provided by Khalifa et al. (2025), the evaluation concentrated primarily on generic object detection tasks. The study did not investigate how the architectures respond to challenges commonly encountered in healthcare datasets, including irregular lesion morphology, image-quality variation, and highly imbalanced disease categories. Consequently, additional studies are needed to assess their applicability in medical image analysis.

Architectural Advancements in YOLO Object Detection Models

Yasir et al. (2025) reviewed the progression of YOLO architectures from earlier versions to more recent releases. The study examined architectural modifications, feature extraction strategies, detection mechanisms, and computational optimizations introduced across successive generations of the framework. Through this review, Yasir

et al. (2025) sought to determine how design innovations contributed to improvements in object detection performance.

The review conducted by Yasir et al. (2025) revealed that each YOLO version introduced enhancements intended to improve detection accuracy, efficiency, or training stability. Architectural innovations such as feature fusion mechanisms, anchor-free detection strategies, and optimized network structures were found to contribute to stronger benchmark performance. However, the authors also emphasized that performance gains often depended on the characteristics of the dataset and application domain.

Although Yasir et al. (2025) provided a comprehensive overview of YOLO's architectural evolution, the review focused primarily on developments within the YOLO family itself. Comparisons with alternative detection architectures were limited, and issues such as explainability, fairness, and robustness under challenging imaging conditions received relatively little attention. These gaps highlight the need for domain-specific investigations that extend beyond benchmark evaluations.

Recent Developments in YOLO Object Detection Architectures

Hidayatullah et al. (2025) reviewed the evolution of modern YOLO architectures from YOLOv8 through YOLO11. The study examined differences in architectural design, feature extraction approaches, detection strategies, and reported benchmark performance. By comparing multiple generations of the same framework, the researchers sought to identify the innovations that contributed to performance improvements over time.

The review found that newer YOLO versions introduced refinements aimed at improving efficiency, feature representation, and detection capability. While later versions reported gains in selected evaluation metrics, the improvements were often incremental rather than transformational. The findings suggest that architectural development remains an ongoing process in which newer versions address specific limitations rather than universally outperforming their predecessors.

A notable limitation of the review is its emphasis on architectural comparisons within a single family of models. Broader comparisons with alternative object detection frameworks were limited, making it difficult to assess whether the observed improvements represent overall advances in object detection or improvements specific to the YOLO ecosystem. Additionally, the review did not explore performance under healthcare-specific conditions, leaving questions regarding clinical applicability unresolved.

Trade-offs Between Detection Accuracy and Processing Speed

Maqsood et al. (2025) investigated the relationship between detection accuracy and computational efficiency by comparing YOLOv8, Faster R-CNN, and SSD. The study recognized that practical deployment often requires balancing performance and resource consumption rather than maximizing a single evaluation metric. To support this objective, the researchers evaluated each architecture using precision, recall, mean Average Precision (mAP), and inference speed.

The results revealed that the three models exhibited different performance profiles. Faster R-CNN achieved strong detection accuracy but incurred greater computational cost. The SSD offered faster processing but showed limitations in detection performance. Meanwhile, YOLOv8 demonstrated a balance between speed

and accuracy. These findings reinforce the idea that architecture selection should depend on operational requirements and deployment constraints.

While the study effectively highlights performance trade-offs, it focuses on a non-medical application domain and does not address issues frequently encountered in healthcare datasets. Factors such as image quality inconsistencies, class imbalance, and demographic variation were not considered. Consequently, the generalizability of these findings to dermatological image analysis remains uncertain and requires further investigation.

Comparison of Convolutional and Transformer-Based Detection Models

Khan et al. (2026) compared convolutional object detection architectures and transformer-based detection models to examine their respective strengths and limitations. The study evaluated YOLOv8, Faster R-CNN, and DETR using benchmark datasets and common performance indicators, including precision, recall, mean Average Precision (mAP), inference speed, and computational complexity. The comparison reflected growing interest in transformer-based approaches within computer vision research.

The findings showed that the evaluated models exhibited different strengths depending on the performance metric considered. DETR demonstrated strong object representation capabilities but required greater computational resources and longer training times. Faster R-CNN maintained competitive localization performance, while YOLOv8 offered efficient real-time detection. These results indicate that architectural choice involves balancing accuracy, efficiency, and computational cost rather than prioritizing a single criterion.

Although the study provides valuable comparisons between convolutional and transformer-based approaches, it does not address challenges specific to medical image analysis. The evaluation focused on general object detection benchmarks and excluded considerations such as interpretability, fairness, and robustness across diverse patient populations. These limitations suggest opportunities for future research investigating how different detection paradigms perform in healthcare applications where transparency and reliability are essential.

Advancements in the YOLO Family of Object Detection Architectures

Hussain (2024) conducted a comprehensive review examining the evolution of the YOLO family of object detection models, with particular emphasis on YOLOv5, YOLOv8, and YOLOv10. The review synthesized recent developments in object detection by comparing architectural improvements, computational efficiency, localization performance, and suitability for real-time computer vision applications across multiple domains, including autonomous systems, industrial automation, surveillance, and healthcare.

The review found that each successive YOLO version introduced architectural refinements that enhanced detection performance while maintaining real-time processing capability. YOLOv5 improved feature extraction through the CSPDarknet backbone, Path Aggregation Network (PANet), and Mosaic data augmentation, resulting in better accuracy and computational efficiency. YOLOv8 further enhanced performance by adopting an anchor-free detection mechanism and a decoupled detection head, reducing model complexity while improving localization accuracy. Meanwhile, YOLOv10 introduced end-to-end non-maximum suppression (NMS)-free training, spatial-channel decoupled downsampling, and large-kernel convolution, significantly reducing inference

latency while preserving competitive detection accuracy. Collectively, these advancements demonstrate that the YOLO family has continuously evolved toward faster and more computationally efficient object detection.

As a review-based study, however, the paper primarily emphasized architectural and computational improvements rather than evaluating performance within specialized application domains. It provided limited discussion on how YOLO models are affected by domain-specific challenges such as illumination variability, melanin-rich skin characteristics, and severe class imbalance commonly encountered in dermatological image analysis. These limitations highlight the need for empirical studies investigating how preprocessing techniques, such as L*-CLAHE luminance enhancement, and imbalance-aware optimization methods, such as Focal Loss, can improve the robustness of modern YOLO architectures for skin lesion localization on diverse skin-tone datasets.

Historical Evolution of YOLO-Based Real-Time Object Detection Systems

Wang and Liao (2024) examined the evolution of the YOLO family of object detection models from YOLOv1 to YOLOv10. The study reviewed the major architectural developments introduced across successive YOLO versions and compared their detection accuracy, computational efficiency, inference speed, and applicability to real-time computer vision tasks. The survey also discussed the growing adoption of YOLO in various domains, including intelligent transportation, robotics, surveillance, manufacturing, and medical image analysis.

The review found that the YOLO family has consistently evolved toward improving computational efficiency while maintaining competitive localization performance. Earlier versions focused on establishing fast single-stage object detection, whereas later versions introduced architectural innovations such as improved backbone

networks, feature pyramid structures, anchor-free detection mechanisms, decoupled detection heads, and end-to-end optimization strategies. These advancements enabled newer YOLO models to achieve higher detection accuracy, faster inference, and better scalability for edge computing and real-time applications. The study concluded that the continuous evolution of the YOLO architecture has established it as one of the most efficient object detection frameworks in modern computer vision.

As the paper primarily emphasized architectural and computational advancements across YOLO versions without experimentally evaluating their robustness in domain-specific applications. In particular, it provided limited discussion on the effects of illumination variability, melanin-rich skin characteristics, and severe class imbalance on object detection performance in dermatological datasets. These limitations indicate the need for application-specific investigations that combine modern YOLO architectures with preprocessing and optimization techniques, such as L*-CLAHE luminance enhancement and Focal Loss, to improve lesion localization performance on diverse skin-tone datasets.

Recent Advancements in the Ultralytics YOLO Architecture

Sapkota and Karkee (2026) presented a comprehensive review of the evolution of the Ultralytics YOLO family, focusing on the architectural developments of YOLOv5, YOLOv8, YOLOv11, and the latest YOLOv26. The study examined the progression of these models by comparing their network architectures, computational efficiency, inference speed, localization performance, and suitability for modern computer vision and pattern recognition applications. The review synthesized recent developments to demonstrate how successive YOLO versions have continuously improved real-time

object detection while maintaining high detection accuracy across diverse application domains.

The review found that each new YOLO generation introduced architectural refinements aimed at increasing computational efficiency and detection performance. YOLOv5 established a strong foundation through improved feature extraction and optimized training strategies, while YOLOv8 adopted an anchor-free detection framework and decoupled detection head to enhance localization accuracy. YOLO11 further optimized feature aggregation and backbone efficiency, resulting in faster convergence and improved parameter utilization. Building upon these developments, YOLOv26 introduced additional network optimizations, lightweight architectural components, and more efficient feature representation techniques, enabling lower inference latency and greater scalability for edge computing and real-time artificial intelligence applications. These findings demonstrate that the evolution of the Ultralytics YOLO family has primarily focused on improving computational performance while preserving detection accuracy.

As a review-based investigation, however, the study mainly evaluated architectural advancements using general computer vision benchmarks and did not experimentally assess the robustness of YOLOv26 in specialized medical imaging environments. The review also provided limited discussion on domain-specific challenges such as illumination variability, melanin-rich skin characteristics, and severe class imbalance that commonly affect dermatological image analysis. These limitations highlight the need for empirical research investigating whether integrating image enhancement techniques, such as L*-CLAHE luminance normalization, together with imbalance-aware optimization methods like Focal Loss, can improve the localization

performance of YOLOv26 on diverse skin-tone datasets representative of the Philippine population.

Recent Developments in Dermatology Foundation Models

Yan et al. (2025) introduced PanDerm, a multimodal vision foundation model developed for clinical dermatology. The study sought to establish a general-purpose dermatological artificial intelligence system capable of supporting multiple clinical tasks, including skin cancer screening, lesion segmentation, risk stratification, differential diagnosis, longitudinal monitoring, and prognosis prediction. PanDerm was pretrained using self-supervised learning on more than two million dermatological images collected from multiple clinical institutions and imaging modalities. The model was subsequently evaluated across numerous dermatological benchmarks to examine its performance and clinical applicability.

The findings demonstrated that PanDerm achieved state-of-the-art performance across multiple dermatological tasks while requiring substantially fewer labeled training samples than conventional deep learning models. The foundation model also improved diagnostic accuracy and demonstrated strong generalization across diverse clinical scenarios, highlighting the potential of large-scale pretraining for dermatological artificial intelligence. These results indicate that foundation models can effectively learn transferable dermatological representations applicable to a wide range of clinical applications.

Despite its strong performance, the study primarily focused on developing a general-purpose dermatology foundation model rather than a specialized object detection framework. The evaluation emphasized broad dermatological capabilities,

including classification, segmentation, and clinical decision support, without specifically investigating object detection challenges such as illumination-induced feature degradation, domain shift across diverse Fitzpatrick skin types, or long-tail class imbalance in infectious skin lesion localization. These limitations suggest the need for specialized object detection frameworks that integrate luminance normalization, decoupled training strategies, and imbalance-aware optimization techniques to improve lesion localization performance in dermatological datasets.

Foundation Models for Zero-Shot Dermatological Image Analysis

Yu et al. (2026) presented DermFM-Zero, a vision-language dermatology foundation model designed to perform zero-shot and low-data dermatological image analysis. The study investigated whether large-scale pretrained foundation models could generalize to unseen dermatological conditions without requiring extensive task-specific training. Using diverse dermatological datasets, the researchers evaluated the model across multiple clinical tasks, including disease classification, rare disease recognition, and low-data learning scenarios, to assess its adaptability and generalization capability.

The findings showed that DermFM-Zero achieved competitive performance across several dermatological benchmarks, particularly in zero-shot classification, few-shot learning, and recognition of rare skin diseases. By leveraging large-scale pretraining and vision-language learning, the model demonstrated strong generalization even when limited labeled data were available. These findings highlight the growing importance of foundation models in supporting flexible and scalable dermatological artificial intelligence systems.

However, the study primarily emphasized vision-language representation learning and generalized dermatological reasoning rather than specialized object detection. The evaluation did not investigate lesion localization using bounding boxes, nor did it examine algorithmic techniques for mitigating illumination-induced feature degradation, domain shift across diverse Fitzpatrick skin types, or long-tail class imbalance during object detection. Consequently, further research is needed to evaluate specialized object detection frameworks that integrate image enhancement, transfer learning, decoupled training strategies, and Focal Loss for infectious skin lesion localization.

The Domain Shift Problem & Melanin-Rich Skin Bias

Dataset Diversity and Skin Tone Representation in Dermatology AI

Alipour et al. (2024) examined the representation of skin tones in publicly available skin lesion datasets used for artificial intelligence applications. The researchers aimed to determine whether existing dermatological datasets adequately represent different skin tones, particularly individuals with darker skin. Their study highlighted concerns regarding dataset imbalance and its potential impact on the fairness and reliability of AI systems.

The researchers evaluated the proportion of images representing different Fitzpatrick skin types and assessed the availability of metadata related to skin pigmentation. Their analysis focused on identifying gaps in skin tone diversity across datasets.

The findings revealed that darker skin tones were significantly underrepresented in many dermatology datasets. Alipour et al. (2024) reported that most datasets predominantly contained images from lighter-skinned populations, resulting in an

imbalance that could affect model generalization. The researchers emphasized that such limitations may contribute to reduced diagnostic performance when AI systems are applied to diverse patient populations.

This study is relevant to the present research for it highlights the importance of developing skin disease detection systems using datasets that include a wide range of Filipino skin tones. The findings of Alipour et al. (2024) support the need to address skin tone bias and improve model fairness.

Bias in Deep Learning-Based Skin Lesion Analysis

Benčević et al. (2024) investigated the presence of skin color bias in deep learning-based skin lesion segmentation systems. The researchers sought to determine whether variations in skin pigmentation influence the performance of computer vision models used in dermatological image analysis. The said study focused on evaluating fairness and robustness across different skin tone categories.

The researchers conducted experiments using skin lesion datasets containing images from multiple skin tone groups. Benčević et al. (2024) compared segmentation performance across lighter and darker skin categories by analyzing metrics such as segmentation accuracy, sensitivity, and overlap scores. This methodology enabled the identification of disparities in model performance.

Results showed that segmentation models generally achieved lower performance on darker skin tones than on lighter skin tones. Benčević et al. (2024) found that limited representation of melanin-rich skin in training datasets contributed to reduced model accuracy. The study emphasized that dataset imbalance can lead to biased predictions and unequal clinical outcomes.

The findings are relevant to the current study because they demonstrate the importance of ensuring equitable AI performance across diverse skin tones. Since the proposed framework targets Filipino patients with varying skin pigmentation, the results of Benčević et al. (2024) support efforts to reduce skin tone bias during model development and evaluation.

Artificial Intelligence Performance on Darker Skin Tones

Dowie et al. (2025) explored the diagnostic capabilities of artificial intelligence systems in dermatology for individuals with darker skin tones. The study aimed to evaluate whether AI models developed using existing datasets could accurately identify dermatological conditions in populations with increased skin pigmentation. The researchers emphasized the need for fairness in medical AI applications.

To accomplish this objective, Dowie et al. (2025) reviewed published studies involving dermatological AI systems and assessed the reported performance across different skin tone groups. The researchers examined datasets, evaluation methodologies, and diagnostic outcomes to identify patterns related to skin tone representation.

The review found that many dermatology AI systems exhibited reduced diagnostic accuracy when applied to darker skin tones. Dowie et al. (2025) attributed these limitations to insufficient representation of melanin-rich skin within training datasets. The researchers concluded that improving dataset diversity is necessary to enhance model generalizability and reduce disparities in diagnostic outcomes. This study supports the present research as it provides evidence that skin tone diversity should be considered when developing AI-based skin disease detection systems.

Domain Shift in Medical Image Analysis

Willemink et al. (2024) investigated the challenges associated with domain shift in medical artificial intelligence systems. The study focused on understanding how models trained on one dataset may experience performance degradation when applied to data from different populations, imaging conditions, or healthcare settings. The researchers highlighted domain shift as a major obstacle to clinical deployment.

The researchers analyzed multiple medical imaging studies involving cross-institutional and cross-population evaluations. Willemink et al. (2024) examined factors contributing to domain shift, including variations in image acquisition devices, demographic characteristics, and environmental conditions. Their review emphasized the importance of external validation in AI research.

The findings indicated that AI models frequently experience reduced accuracy when evaluated on datasets that differ from the training data. Willemink et al. (2024) reported that demographic differences, including skin tone variation, can significantly influence model performance. The study emphasized the need for robust preprocessing techniques and diverse datasets to mitigate domain shift.

Color Space Transformations (L^* channel vs RGB)

Color Space Selection for Deep Learning-Based Skin Analysis

Chen et al. (2025) examined the impact of different color spaces on deep learning performance for skin image analysis. The study focused on determining whether alternative color representations could improve the detection and classification of skin lesions. The researchers noted that color information plays a critical role in dermatological diagnosis.

To investigate this issue, Chen et al. (2025) trained deep learning models using skin lesion images represented in RGB, HSV, and CIELAB color spaces. The models were evaluated based on classification accuracy, sensitivity, specificity, and feature extraction quality. This approach enabled the researchers to compare the influence of color representation on model performance.

The findings revealed that models trained using CIELAB-transformed images achieved better overall performance than those trained using raw RGB images. Chen et al. (2025) attributed this improvement to the separation of luminance and chromatic information, which allowed more discriminative features to be extracted from skin lesions. The researchers highlighted the usefulness of the L^* channel for emphasizing lesion structures. The study is relevant to the present research as it demonstrates the potential benefits of using CIELAB color transformations in dermatological image analysis.

Image Enhancement Techniques for Healthcare Imaging Systems

Dela Cruz et al. (2025) investigated image enhancement techniques for healthcare imaging applications in the Philippines. The study aimed to improve image quality obtained from low-cost imaging devices commonly used in local healthcare settings. The researchers focused on preprocessing methods that could enhance visual information while preserving diagnostic details.

The study evaluated several image enhancement techniques, including histogram equalization, CLAHE, and color space transformations. Dela Cruz et al. (2025) assessed image quality through contrast analysis and expert evaluation. The researchers emphasized the importance of preprocessing for improving automated image analysis systems.

The findings indicated that color space transformation combined with CLAHE produced superior image quality compared to enhancement performed directly in RGB space. Dela Cruz et al. (2025) observed improvements in contrast, feature visibility, and image consistency. These enhancements were particularly beneficial for images captured under non-standard lighting conditions. This study is relevant to the current research because smartphone-acquired skin disease images may exhibit varying illumination and contrast levels.

CLAHE-Based Image Enhancement for Medical Image Classification

Halloum et al. (2025) investigated the impact of image enhancement techniques on medical image classification performance. The researchers emphasized that image quality plays a crucial role in deep learning systems because poor contrast and uneven illumination can obscure important visual features. Their study explored the use of Contrast Limited Adaptive Histogram Equalization (CLAHE) as a preprocessing technique to improve the visibility of relevant image characteristics before classification.

The study found that CLAHE effectively enhanced local contrast while preventing excessive amplification of noise. Unlike traditional histogram equalization, CLAHE operates on small image regions and applies contrast limiting, resulting in clearer image details without introducing significant distortion. This capability makes CLAHE particularly suitable for medical imaging applications where subtle visual differences are critical for diagnosis.

Furthermore, the researchers observed that enhanced image quality contributed to improved feature extraction by deep learning models. By increasing the visibility of important structures and boundaries, CLAHE enabled neural networks to learn more discriminative features from the input images. This enhancement process ultimately

supported more reliable image classification performance across different medical imaging datasets.

The findings of Halloum et al. (2025) support the use of CLAHE in the proposed study because the technique can improve lesion visibility while preserving essential image information. Applying CLAHE to the lightness component of skin disease images may help the YOLO model detect disease-related patterns more effectively, particularly when images are affected by inconsistent lighting conditions.

Utilizing L^ , a^* , and b^* Channels in Skin Disease Detection*

Al-Hatab et al. (2025) examined the effectiveness of the CIELAB color space for analyzing skin lesion images. The researchers highlighted that CIELAB separates brightness information from color information, allowing image processing techniques to target specific visual properties without affecting the entire image. This characteristic makes CIELAB particularly useful for dermatological image analysis.

The study explained that the L^* channel represents lightness, while the a^* and b^* channels represent color information along the red-green and blue-yellow axes, respectively. By separating these components, the researchers were able to analyze lesion characteristics more effectively. They noted that skin lesions often exhibit distinctive color variations that can serve as important indicators during disease identification and classification.

The results showed that color-based analysis using the a^* and b^* channels provided valuable information regarding lesion appearance and pigmentation patterns. Meanwhile, the L^* channel contributed information related to brightness and texture

variations. The combination of these channels allowed for a more comprehensive representation of skin lesion characteristics compared with conventional RGB images.

This study is relevant to the present research because it demonstrates the importance of preserving the original a^* and b^* chromatic information while enhancing only the L^* channel. This preprocessing strategy supports more accurate localization and classification of infectious skin lesions across diverse Fitzpatrick skin types.

Transfer Learning & Focal Loss on Imbalanced Data

Addressing Class Imbalance Using Focal Loss

Lin et al. (2024) examined the effectiveness of focal loss in addressing class imbalance problems in object detection systems. The researchers noted that traditional loss functions often struggle when datasets contain significantly more samples from one class than another. Their study focused on improving model performance for minority classes without sacrificing overall accuracy.

The researchers implemented focal loss within object detection architectures and compared its performance with conventional cross-entropy loss functions. Lin et al. (2024) evaluated detection performance using precision, recall, and mean Average Precision (mAP) across balanced and imbalanced datasets. This approach enabled a direct comparison of loss function effectiveness.

The findings revealed that focal loss significantly improved the detection of minority classes by reducing the influence of easy negative samples during training. Lin et al. (2024) reported that RetinaNet equipped with focal loss achieved up to 39.1 AP on the COCO test-dev benchmark and 37.8 AP using a ResNet-101-FPN backbone, outperforming many existing one-stage detectors while remaining competitive with

two-stage approaches. The researchers observed that focal loss improved the model's ability to focus on hard and underrepresented examples, leading to stronger detection performance and enhanced sensitivity for minority classes. Consequently, Lin et al. (2024) concluded that focal loss provides an effective and robust solution to class imbalance challenges in object detection and computer vision applications.

Transfer Learning in Dermatological Image Analysis

Rahman et al. (2025) investigated transfer learning techniques for automated skin disease classification. The study aimed to evaluate whether pre-trained deep learning models could improve the recognition of dermatological conditions across diverse image datasets. The researchers highlighted the importance of robust feature extraction in skin lesion analysis.

To perform the study, Rahman et al. (2025) fine-tuned several pre-trained neural networks using dermatological image datasets. Performance was assessed through accuracy, sensitivity, specificity, and F1-score. The researchers compared transfer learning approaches with conventional training methods to determine their effectiveness.

The findings indicated that transfer learning consistently improved classification performance and reduced training requirements. Rahman et al. (2025) observed that pre-trained models were able to extract more discriminative lesion features than models trained from scratch. These improvements resulted in more reliable disease recognition. This supports the use of pre-trained weights to improve model generalization and diagnostic performance across diverse skin disease categories.

Loss Functions for Imbalanced Medical Image Classification

Zhao et al. (2025) explored the use of focal loss in medical image classification tasks involving highly imbalanced datasets. The researchers aimed to determine whether focal loss could improve recognition of rare disease categories while maintaining overall model performance. The study addressed a common challenge in medical artificial intelligence applications.

The researchers trained deep learning models using both focal loss and standard cross-entropy loss functions. Zhao et al. (2025) evaluated performance using sensitivity, specificity, precision, and recall. The experiments focused on assessing how effectively each loss function handled underrepresented disease classes.

The results demonstrated that focal loss improved minority class detection and reduced classification bias toward majority classes. Zhao et al. (2025) found that the loss function enabled models to focus on difficult examples during training, leading to improved diagnostic performance. The researchers concluded that focal loss is particularly beneficial for medical datasets with severe class imbalance.

Decoupled Training for Long-Tailed Medical Image Classification

Ju et al. (2024) introduced MONICA, a comprehensive benchmark for evaluating long-tailed medical image classification (LTMIC) systems. The study highlighted that real-world medical datasets naturally exhibit long-tailed distributions, where a few common diseases dominate the dataset, and rare conditions are severely underrepresented. The researchers found that training a model end-to-end on such imbalanced data causes the classifier's weight norms to become heavily biased toward majority classes, leading to the systematic underprediction of rare diseases.

To address this, Ju et al. (2024) identified two-stage decoupled training as a highly effective general paradigm. In the first stage (representation learning), the model's feature encoder is trained on the entire imbalanced dataset, allowing it to learn foundational, generalized visual features (such as textures, edges, and shapes) without discarding valuable data. In the second stage (classifier re-training), the feature encoder is frozen, and only the classifier head is fine-tuned using re-balancing strategies. The study demonstrated that decoupling these processes prevents feature degradation while significantly improving the model's ability to detect minority classes.

These findings directly support the methodology of the present study. By adopting a two-stage decoupled training pipeline, the proposed framework can preserve the natural long-tailed distribution of Philippine dermatological datasets. The YOLO backbone can be utilized for generalized representation learning across all available melanin-rich samples, while Focal Loss can be applied exclusively during the classifier re-training stage to mitigate bias against underrepresented skin conditions.

Explainable AI (Grad-CAM)

Explainable Artificial Intelligence in Medical Imaging

Kinger and Kulkarni (2024) Explainable Artificial Intelligence (XAI) techniques in medical imaging to address the growing need for transparency and interpretability in deep learning-based healthcare systems. The researchers emphasized that although artificial intelligence models have demonstrated strong diagnostic capabilities, their black-box nature remains a major barrier to clinical acceptance. The study aimed to examine how explainability methods can improve trust, accountability, and decision-making in medical image analysis.

They analyzed various XAI techniques commonly applied in medical imaging, including Gradient-weighted Class Activation Mapping (Grad-CAM), Saliency Maps, Local Interpretable Model-Agnostic Explanations (LIME), and other visualization-based approaches. The researchers evaluated how these techniques provide explanations for model predictions and assist healthcare professionals in identifying clinically relevant image regions. Their analysis focused on the effectiveness, interpretability, and practical applicability of different XAI methods across medical imaging tasks.

The findings showed that Grad-CAM was among the most widely adopted XAI techniques due to its ability to generate intuitive heatmaps highlighting the image regions that contribute most to model predictions. Kinger and Kulkarni (2024) reported that Grad-CAM provides more localized and clinically interpretable visual explanations than methods such as Saliency Maps and LIME, which may produce noisy or less consistent results. The researchers further noted that Grad-CAM effectively helps clinicians understand model decisions, thereby improving transparency, trustworthiness, and confidence in AI-assisted medical diagnosis.

Grad-CAM for Skin Lesion Classification

Patel et al. (2024) examined the effectiveness of Grad-CAM in explaining deep learning predictions for skin lesion classification. The researchers sought to determine whether Grad-CAM could accurately highlight lesion regions used by convolutional neural networks during diagnosis. Their study addressed concerns regarding transparency in AI-based dermatology systems.

The researchers trained deep learning models on skin lesion image datasets and generated Grad-CAM visualizations for each prediction. Patel et al. (2024) compared the generated heatmaps with dermatologist annotations to evaluate localization accuracy.

This approach enabled the researchers to assess whether the model focused on medically relevant lesion features.

The results indicated that Grad-CAM successfully highlighted lesion structures associated with diagnostic decisions. Patel et al. (2024) found that the heatmaps corresponded closely with clinically important regions identified by dermatologists. These findings demonstrated that Grad-CAM can provide meaningful explanations for deep learning predictions in dermatological applications.

Explainable Deep Learning for Clinical Decision Support

Rahman et al. (2025) investigated explainable deep learning techniques for clinical decision support systems. The researchers aimed to determine whether visual explanation methods could improve the interpretability of AI-generated medical diagnoses. The study highlighted the importance of transparency in healthcare environments where decisions may directly affect patient outcomes.

To conduct the study, Rahman et al. (2025) implemented Grad-CAM within deep learning classification models and evaluated the resulting visual explanations using expert review. Medical professionals assessed whether the highlighted regions corresponded to clinically meaningful features. This evaluation enabled the researchers to measure the practical usefulness of explainability methods.

The findings showed that Grad-CAM improved clinician understanding of model predictions and increased confidence in automated diagnostic systems. Rahman et al. (2025) reported that healthcare professionals preferred models that provided visual explanations compared to black-box models. The study concluded that explainability can facilitate the adoption of AI technologies in clinical settings.

Digital Morphology of the Target Infectious Skin Diseases

Morphological Characteristics of Bacterial and Fungal Skin Diseases

Chauhan et al. (2023) conducted a systematic review of the dermoscopic and morphological features of bacterial and fungal skin infections. The study aimed to identify characteristic visual patterns that assist in differentiating infectious skin diseases and support accurate dermatological diagnosis. The researchers emphasized the importance of lesion appearance, color variation, surface texture, scaling, and border characteristics in distinguishing different types of skin infections.

The review examined several bacterial skin diseases, including Impetigo, Ecthyma, Folliculitis, Erythrasma, and cellulitis-related conditions, as well as fungal infections such as Tinea Corporis (Ringworm), Tinea Versicolor, Tinea Capitis, and Onychomycosis. Chauhan et al. (2023) reported that bacterial infections commonly manifest as erythematous lesions with pustules, yellowish scales, honey-colored crusts, and irregular borders. In contrast, fungal infections frequently present annular or irregularly shaped lesions with well-defined borders, scaling, and characteristic texture changes such as dry, flaky, or scaly skin.

The findings revealed distinct morphological differences between bacterial and fungal skin diseases. Chauhan et al. (2023) noted that Impetigo often appears with honey-colored crusts, while Folliculitis is characterized by pustules surrounding hair follicles. Meanwhile, Tinea Corporis typically presents ring-shaped lesions with central clearing and elevated erythematous borders, whereas Tinea Versicolor commonly appears as irregular hypo- or hyperpigmented patches with fine scaling. These visual characteristics provide valuable information regarding lesion shape, texture, color, and boundary definition, making them suitable features for image-based disease analysis.

Morphological Features of Viral Skin Diseases

Rahim et al. (2025) investigated the visual characteristics of viral skin lesions through the analysis of Mpox images using deep learning and explainable artificial intelligence (XAI) techniques. The researchers aimed to identify lesion features that could support the early detection of viral skin diseases and improve automated image-based diagnosis. Their study emphasized the importance of lesion morphology in distinguishing viral infections from other dermatological conditions.

The study examined Mpox skin lesion images and analyzed their visual patterns using pretrained convolutional neural network (CNN) models. Rahim et al. (2025) observed that viral lesions commonly appear as vesicles, pustules, and crusted lesions that progress through distinct developmental stages. The researchers noted that these lesions often exhibit characteristic shapes, textures, and spatial distributions that can be recognized through image analysis.

The findings revealed that Mpox lesions display noticeable morphological variations, including fluid-filled vesicles, elevated pustules, and crusted surfaces during later stages of infection. Rahim et al. (2025) reported that lesion progression creates distinctive visual patterns that can be effectively identified by deep learning models. The study demonstrated that explainable AI techniques can highlight lesion regions and morphological features that contribute to classification decisions.

Morphological Overlap in Dermatology

Fine-Grained Visual Categorization (FGVC) of the Target Infectious Diseases

Fine-Grained Visual Categorization (FGVC) remains a persistent challenge in dermatological image analysis because infectious skin diseases can exhibit overlapping

morphological characteristics. The eight conditions covered in the present study— (1) *four viral*: Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, and Warts (*Verruca(Kulugo)*), (2) *three fungal*: Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), and Athlete’s Foot (*Tinea Pedis (Alipunga)*), and (3) *one bacterial*: Impetigo (*Impetigo Contagiosa (Mamaso)*)— may present with visually similar features that complicate differentiation (Chauhan et al., 2023; Hussain et al., 2022). Viral conditions such as Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), and Hand, Foot, and Mouth Disease may present with papular or vesicular lesions, while Warts (*Verruca(Kulugo)*) may exhibit raised or hyperkeratotic surfaces (Chauhan et al., 2023; Hussain et al., 2022). Fungal infections such as Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), and Athlete’s Foot (*Tinea Pedis (Alipunga)*) may involve scaling, erythema, pigmentation changes, and distinct lesion boundaries, whereas Impetigo (*Impetigo Contagiosa (Mamaso)*) may present with vesicles, pustules, erosions, and crusting (Chauhan et al., 2023; Leung et al., 2022). These clusters are utilized strictly as analytical groupings for fine-grained computer vision evaluation, not as official medical or diagnostic classifications.

Accurate differentiation therefore requires attention to fine-grained features such as lesion shape, texture, scaling, crusting, pigmentation, and boundaries rather than broad visual appearance alone (Chauhan et al., 2023). This challenge is further complicated by skin-tone variation, as dermatological AI models have demonstrated performance disparities on darker skin tones (Daneshjou et al., 2022). These findings highlight the need for approaches that preserve clinically relevant visual information while enabling models to learn subtle disease-specific morphological characteristics across diverse skin tones.

Transfer Learning for Skin Disease Classification in the Philippine Context

Velasco et al. (2023) investigated the application of transfer learning and convolutional neural networks (CNNs) for automated skin disease classification in the Philippine context. The researchers developed a dataset containing 3,400 dermatological images representing seven diseases prevalent in the Philippines, namely chickenpox, acne, eczema, pityriasis rosea, psoriasis, tinea corporis, and vitiligo. Several pretrained deep learning models were evaluated to determine their ability to recognize visual patterns associated with these conditions. Among the evaluated models, MobileNet achieved the highest classification accuracy, demonstrating the potential of transfer learning to adapt pretrained visual representations to locally relevant dermatological images.

They demonstrate that pretrained deep learning models can be effectively adapted to a Philippine dermatological dataset despite the differences between general-purpose training data and medical images. The study also highlights the potential of locally relevant image datasets for developing automated skin-disease classification systems. According to Velasco et al. (2023), the inclusion of diseases such as chickenpox and tinea corporis alongside non-infectious conditions provides a diverse set of visual patterns for evaluating automated classification. This is important because practical dermatological systems may encounter diseases with different appearances, lesion structures, and levels of visual similarity.

Building upon the Philippine-based findings of Velasco et al. (2023), the present study applies transfer learning through YOLOv26 while extending the task from image-level classification toward lesion localization and disease classification.

Automated Skin-Lesion Segmentation in the Philippine Context

The study recognized that skin-lesion images may contain variations in image quality, illumination, background appearance, and visual artifacts, which can interfere with accurate lesion-boundary identification. To address this challenge, Salido et al. (2025/2026) applied semantic segmentation to separate affected regions from surrounding skin, producing a more focused representation of the lesion for subsequent analysis.

The findings of Salido et al. (2026) demonstrate the importance of accurate lesion-region separation in computer-assisted dermatological analysis. By identifying the spatial boundaries of a lesion, segmentation can reduce the influence of irrelevant surrounding areas and allow computational models to concentrate on clinically meaningful visual information. This is particularly relevant for dermatological images in which lesions may have irregular boundaries, gradual color transitions, varying shapes, or limited contrast against surrounding skin. Their use of DeepLabv3+ with a ResNet-18 backbone further demonstrates how deep learning can be applied to learn spatial representations from complex dermatological images.

A model that can reliably determine where a lesion is located can reduce the influence of unrelated image regions and provide a more appropriate basis for subsequent classification. This supports the localization component of the proposed framework, where YOLOv26 is used to identify the spatial region of infectious skin lesions before the model focuses on more detailed morphological characteristics.

Feature-Importance Analysis for Skin-Lesion Classification in the Philippine Context

They investigated machine learning-based skin-lesion classification using Support Vector Machines (SVM) and dermoscopic images from the Philippines. The study incorporated feature-importance analysis to determine which image characteristics

contributed to the classification of different skin lesions. The researchers considered several clinically relevant visual attributes, including shape, size, color, and texture, allowing the study to examine the relationship between measurable image features and lesion classification. This approach provided a more feature-oriented perspective on automated dermatological analysis.

The findings emphasized that morphological and visual characteristics contain useful information for distinguishing skin lesions. Shape and size can describe the structural configuration of a lesion, while color and texture provide additional information about its surface and visual appearance. The inclusion of feature-importance analysis also helps identify which characteristics have greater influence on the classification process, providing insight into the visual evidence used by the machine-learning model. These results support the importance of capturing multiple complementary characteristics rather than relying on a single visual attribute when analyzing dermatological lesions.

The study is particularly relevant to the present research because several of the identified characteristics, especially shape and texture, correspond to the detailed morphological information that may be important when diseases have overlapping appearances.

Deep Learning for Differentiating Morphologically Similar Skin Diseases

Eskandari and Sharbatdar (2024) investigated the use of deep learning for the automated diagnosis of psoriasis and lichen planus, two cutaneous diseases that may exhibit similar visual and morphological characteristics. The study aimed to determine

whether deep learning could distinguish between these conditions despite their overlapping clinical appearances. Using dermatological images and classification performance measures, Eskandari and Sharbatdar (2024) evaluated the ability of deep learning models to identify disease-specific patterns that may not be easily distinguished through general visual appearance alone.

The findings of Eskandari and Sharbatdar (2024) demonstrated that deep learning can assist in differentiating visually similar skin diseases by learning relevant visual characteristics from dermatological images. Their results emphasize the importance of identifying subtle disease-specific features when conditions share comparable clinical presentations. This suggests that automated dermatological classification systems may benefit from representations that capture detailed morphological information rather than depending primarily on broad or general visual patterns. The study therefore provides support for the use of feature-learning approaches when dealing with diseases that exhibit substantial morphological similarity.

The relevance of Eskandari and Sharbatdar (2024) to the present study lies in its demonstration that morphological similarity can create a significant challenge for automated skin-disease classification. Consequently, while Eskandari and Sharbatdar (2024) establish the importance of learning disease-specific visual characteristics for morphologically similar conditions, their study leaves an opportunity to investigate whether a more specialized representation-learning strategy can improve discrimination among morphologically similar infectious skin diseases.

Artificial Intelligence and Morphological Features for Differentiating Infectious Skin Lesions

The study focused on determining whether image-derived radiomics features could provide useful information for distinguishing early syphilis from lesions with similar visual presentations. Sun et al. (2025) extracted and analyzed various radiomics characteristics from skin-lesion images, including morphological features, to identify the features that contributed most to classification performance. Their approach demonstrates how quantitative image analysis can provide additional information beyond general visual appearance when differentiating dermatological conditions.

The findings showed that 2D sphericity was among the most predictive features for distinguishing early syphilis from other skin lesions. This result highlights the importance of lesion morphology and shape in infectious skin-disease classification. According to Sun et al. (2025), morphological characteristics can provide discriminative information even when different lesions exhibit visually similar appearances. The findings therefore support the idea that automated systems should capture detailed structural characteristics of lesions rather than relying solely on broad visual patterns such as overall color or appearance.

The work of Sun et al. (2025) is particularly relevant to the present study because it demonstrates the potential of morphological information for distinguishing infectious skin lesions. Features related to shape and structure may help differentiate diseases that share similar clinical appearances, making morphological representation an important consideration in automated infectious skin-disease analysis.

Consequently, the present study extends the direction established by Sun et al. (2025) by incorporating fine-grained morphological representation learning into the classification-focused stage of a YOLOv26 framework. After localization representation learning and selected backbone layer freezing, the proposed framework focuses on

learning detailed characteristics such as texture, crusting, scaling, edges, shape, and lesion boundaries.

Limitation of Static Contrast Enhancement

Adaptive Clip-Limit Selection for Improved Contrast Enhancement

Fawzi et al. (2021) investigated an adaptive histogram equalization method for non-homogenized intensity images, focusing on the limitations of conventional contrast-enhancement techniques that use predetermined parameters. Fawzi et al. (2021) proposed an approach that adaptively determines the clip limit and tile size according to the characteristics of the input image. This approach was designed to provide more appropriate local contrast enhancement across images with different intensity distributions rather than applying the same configuration to every image.

The findings of Fawzi et al. (2021) showed that inappropriate fixed enhancement parameters may produce undesirable visual effects, including artifacts, unnatural appearances, and washed-out regions. Adapting the enhancement parameters to the characteristics of the input image can produce more suitable contrast enhancement and reduce these unwanted effects. This finding is relevant to dermatological image processing because differences in illumination and image intensity may influence the visibility of lesion structures, boundaries, and other important visual characteristics.

It provides an important basis for moving from static contrast enhancement toward adaptive parameter selection. Rather than assuming that a single clip limit and tile size are appropriate for every image, the method considers image-specific characteristics when determining the enhancement configuration. This principle is relevant to the present study because infectious skin-lesion images may vary in

luminance and pigmentation, potentially causing a fixed enhancement setting to produce inconsistent results across different images.

Optimization-Based Parameter Selection for CLAHE

The study addressed the challenge of determining suitable CLAHE parameters for effective image contrast enhancement, recognizing that the resulting image quality can vary depending on the selected configuration. Kuran and Kuran (2021) focused on optimizing parameters such as the clip limit and tile size rather than relying solely on manually selected or fixed values. Their approach aimed to identify parameter combinations that could provide improved contrast enhancement while satisfying multiple enhancement objectives.

The findings of Kuran and Kuran (2021) demonstrated that CLAHE performance is strongly influenced by its parameter configuration. Through the optimization-based approach, the researchers identified parameter combinations that improved contrast enhancement across different image conditions. The results suggest that a parameter setting that works effectively for one image may not necessarily provide the same level of enhancement for another image with different intensity characteristics. Thus, Kuran and Kuran (2021) provide evidence that CLAHE parameter selection can benefit from optimization rather than depending entirely on predetermined values.

Their optimization strategy demonstrates that contrast enhancement can be improved when the parameters are determined according to the characteristics and objectives of the enhancement process. This principle is important for dermatological images because variations in illumination, pigmentation, and image intensity may cause a single enhancement configuration to produce different results across images.

Image-Adaptive Clip-Limit Estimation for CLAHE

The researchers recognized that images may have different histogram distributions and intensity patterns, making a single clip-limit value unsuitable for every input. Otsuka et al. (2026) therefore proposed an image-adaptive approach that estimates the clip limit based on information derived from the input image, with the goal of producing more appropriate contrast enhancement across varying image conditions.

Their approach showed that image-adaptive clip-limit estimation can address the limitations associated with applying a uniform parameter configuration, particularly when images contain substantially different intensity distributions. According to Otsuka et al. (2026), fixed clip limits may contribute to excessive enhancement when the selected value does not correspond appropriately to the input image. This supports the broader principle that contrast-enhancement parameters should be responsive to image characteristics rather than being treated as universally optimal values.

The work of Otsuka et al. (2026) is particularly relevant to the present study because it provides recent evidence supporting the transition from static CLAHE toward image-adaptive clip-limit estimation. Despite its contribution, Otsuka et al. (2026) focused on general image enhancement and did not specifically examine dermatological images, skin pigmentation, melanin-rich skin, or infectious skin lesions.

ITA / Skin Tone Measurement

Individual Typology Angle and Fitzpatrick Skin Phototypes in Skin Characterization

The study clarified that these two measures describe different characteristics of skin. Osto et al. (2022) explained that ITA provides a quantitative measure associated

with constitutive skin pigmentation and measured skin color, whereas the Fitzpatrick system primarily categorizes individuals according to their response to ultraviolet exposure, particularly their tendency to burn or tan. Therefore, Osto et al. (2022) emphasized that ITA values and Fitzpatrick categories should not be treated as interchangeable measures.

The study further demonstrated the usefulness of ITA as a continuous numerical representation of skin pigmentation. Unlike the categorical nature of Fitzpatrick classification, ITA provides a numerical value that can represent variations in skin color more continuously. Osto et al. (2022) described ITA as being derived from the L^* and b^* components of the CIELAB color space, allowing differences in skin lightness to be quantified. This distinction is particularly relevant to image-based dermatological research because individuals belonging to the same Fitzpatrick category may still exhibit variations in measured skin color. Thus, as demonstrated by Osto et al. (2022), ITA can provide an additional computational description of skin characteristics while Fitzpatrick classification can remain useful for categorical grouping and performance analysis.

The findings provide an important theoretical basis for the present study's use of ITA as a computational variable. Since ITA represents skin-lightness characteristics numerically, it can potentially be used as an input factor for image-processing decisions rather than only as a descriptive skin measurement.

Image-Adaptive Clip-Limit Estimation for Skin-Tone-Aware Enhancement

Otsuka et al. (2026) proposed IA-CLAHE (Image-Adaptive Clip Limit Estimation for CLAHE) to address the limitations of conventional CLAHE, particularly the use of a fixed clip limit across images with different histogram distributions. The researchers recognized that a constant clip limit may not provide appropriate enhancement for every

input image because image intensity characteristics can vary considerably. Otsuka et al. (2026) therefore developed an image-adaptive approach that estimates the clip limit according to characteristics extracted from the input image, allowing the enhancement strength to respond to variations in image content.

They demonstrate an important shift from static parameter selection toward adaptive contrast enhancement. Rather than applying one predefined clip limit to all images, their method determines the enhancement parameter based on the characteristics of each input. This approach provides a stronger basis for addressing the limitations of manually selected CLAHE configurations, particularly when images have substantially different intensity distributions. The work of Otsuka et al. (2026) therefore supports the principle that contrast-enhancement parameters should be responsive to the characteristics of individual images.

The study is particularly relevant to the present research because Otsuka et al. (2026) demonstrates that the clip limit itself can be treated as an adaptive variable rather than a fixed configuration. This provides a methodological foundation for investigating whether a more domain-specific characteristic can be used to guide CLAHE parameter selection.

Synthesis of the Study

The reviewed literature demonstrates significant advancements in image enhancement and dermatological AI. However, a critical gap remains in addressing Fine-Grained Visual Categorization (FGVC) on melanin-rich skin. While studies such as Otsuka et al. (2026) have introduced image-adaptive CLAHE (IA-CLAHE) for general

photography, and Osto et al. (2022) established the Individual Typology Angle (ITA) for skin tone measurement, the reviewed literature does not identify a study that connects ITA-based skin tone measurement with adaptive clip-limit selection to address morphological overlap in infectious skin lesion detection. Existing detectors may struggle to distinguish visually similar infectious lesions when relevant morphological features are less distinguishable under low-luminance conditions. The proposed framework directly addresses this gap by engineering a domain-specific $clip_limit = f(ITA)$ logic gate.

The reviewed literature also demonstrates that morphological similarity presents a significant challenge in infectious skin disease classification. Studies of infectious skin conditions have shown that visually similar lesions may require attention to subtle characteristics such as shape, texture, scaling, crusting, pigmentation, and lesion boundaries for accurate differentiation (Chauhan et al., 2023). However, existing studies generally rely on conventional feature extraction, image-level classification, or general deep-learning representations rather than explicitly separating localization learning from subsequent fine-grained morphological representation learning. This creates an opportunity to develop a training strategy that first establishes reliable lesion localization representations and subsequently focuses the model on detailed disease-specific morphology.

Based on these gaps, the present study proposes the ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework for infectious skin lesion localization and classification. The framework integrates CIELAB transformation, ITA estimation, empirically calibrated ITA-guided adaptive L*-CLAHE, YOLOv26 transfer learning, sequential localization representation learning, layer freezing, fine-grained morphological representation learning, Focal Loss, and Grad-CAM. The adaptive enhancement follows the relationship $clip_limit=f(ITA)$, allowing the luminance-processing strength to be

determined according to the skin-lightness characteristics of each input image. Following enhancement, YOLOv26 learns localization representations before selected backbone layers are frozen and the model proceeds to fine-grained learning focused on disease-specific morphological characteristics.

The reviewed literature provides theoretical and empirical support for the individual components of the proposed framework while revealing a gap in their skin-tone-aware and morphology-focused integration for infectious skin diseases. The present study addresses this gap by combining adaptive luminance normalization, lesion localization, sequential representation learning, fine-grained morphological discrimination, imbalance-aware optimization, and explainability within a unified framework. Its effectiveness will be evaluated across Fitzpatrick Skin Types I–V, with particular emphasis on melanin-rich skin (Types III–V), providing a basis for developing a more robust and interpretable approach to infectious skin lesion analysis.

Chapter 3

THE METHODOLOGY

This chapter provides the necessary logical connections between the findings of the past studies and the methodology of the present study.

Research Design

This study will employ an experimental quantitative model-evaluation research design to develop and evaluate the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework for the localization and classification of infectious skin lesions across Fitzpatrick Skin Types I–V, with particular emphasis on melanin-rich skin types III–V.

The study will use a controlled experimental approach in which YOLOv26 will serve as the base object detection architecture, while different preprocessing and training configurations will be evaluated under consistent experimental conditions. The configurations will consist of:

1. Model A: Baseline YOLOv26 (No L*-CLAHE, No Focal Loss)
2. Model B: YOLOv26 + Fixed L*-CLAHE
3. Model C: YOLOv26 + Focal Loss
4. Model D: The Proposed Framework (ITA-Guided Adaptive L*-CLAHE + Focal Loss + Decoupled Training)

The four configurations will be evaluated using the same dataset, disease categories, image resolution, annotation procedures, dataset partitioning, training environment, hardware, and evaluation protocol. This controlled comparison is intended to ensure that observed differences in model performance are primarily attributable to

the computational configurations being investigated. The revised framework specifically identifies CIELAB transformation, ITA estimation, adaptive L*-CLAHE, YOLOv26 transfer learning, Focal Loss optimization, sequential training, and fine-grained morphological representation learning as its major computational components.¹¹¹

The global foundation dataset will be divided into training, validation, and testing subsets using a stratified 70/20/10 split, while maintaining the intended representation of disease categories and Fitzpatrick Skin Types, with the clinical hold-out set kept strictly separate for out-of-distribution testing. The model will be trained using YOLOv26 transfer learning. For the proposed configuration, the training process will use a two-stage procedure in which the first stage focuses on lesion localization representations and the succeeding stage focuses on fine-grained morphological representation learning.

For the adaptive preprocessing configuration, images will undergo CIELAB transformation and ITA estimation. The calculated ITA value will determine the CLAHE clipping parameter through the proposed adaptive relationship before the enhanced image is reconstructed and processed by YOLOv26. This allows the proposed configuration to be evaluated against the unprocessed baseline, the fixed-parameter enhancement configuration, and the Focal Loss-only configuration.

The performance of the four configurations will be evaluated using mAP₅₀, mAP_{50–95}, precision, recall, and F1-score. Performance will also be examined across Fitzpatrick Skin Types I–V, with particular attention to the difference between lighter skin types (I–II) and melanin-rich skin types (III–V).

The resulting performance measurements will be subjected to appropriate descriptive and inferential statistical analysis to determine whether significant differences

exist among the experimental configurations and whether performance improvement varies across the specified skin-tone groups.

Sources of Data

The data used in this study will be drawn from two strategically combined sources, constituting a hybrid dataset designed to provide both sufficient training volume and a real-world domain shift stress test using the genuinely local Philippine subset, focused on eight morphologically overlapping infectious skin diseases within Fitzpatrick Skin Types I-V.

The global foundation (comprising approximately 90–95% of the total dataset) will consist of skin lesion images sourced from publicly available dermatological repositories, specifically DermNet and Kaggle, providing representation across Fitzpatrick Skin Types I–V. These repositories were selected based on their established use in dermatological AI research, the volume and variety of skin lesion images they provide, and (critically) the availability of Fitzpatrick skin type metadata or sufficient visual information to enable skin tone verification during the curation process (Alipour et al., 2024; Groh et al., 2021). Preliminary source-availability scouting across DermNet and supplementary repositories indicates strong existing coverage for four of the eight target diseases — Warts (*Verruca(Kulugo)*), Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), and Hand, Foot, and Mouth Disease in particular show substantial indexed image counts well above the study's minimum threshold, while the remaining four diseases Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Athlete's Foot (*Tinea Pedis (Alipunga)*), and Impetigo (*Impetigo Contagiosa (Mamaso)*), have partial existing coverage that is either close to or below the minimum threshold and will require additional sourcing beyond the primary repositories to reach the target range. These

datasets serve as the foundational resource for pre-training, validation, and comparative evaluation purposes throughout the study.

The clinical hold-out set (comprising approximately 5–10% of the total dataset, an estimated 120 to 400 images distributed across the eight infectious skin disease subtypes within the study's scope, approximately 15 to 50 images per disease) will consist of locally sourced Philippine clinical images collected from consenting Filipino respondents through a structured Google Form instrument, supplemented by global repository data where necessary. Respondents must be at least 18 years of age, have a doctor-confirmed diagnosis of one of the eight target infectious skin diseases, and may belong to any image-based proxy Fitzpatrick Skin Type I–V, with recruitment prioritized toward Fitzpatrick Skin Types III–V given the study's melanin-rich analytical focus. Through the Google Form, respondents will provide their confirmed diagnosis and upload a photograph of their specific skin disease. The questionnaire instrument will be validated by at least three Registered Psychometricians (RPMs) prior to deployment. The study will undergo ethics clearance from PUP Sta. Mesa before any data collection from respondents takes place. In the event that the structured Google Form yields an insufficient number of qualifying respondent submissions across the eight target disease subtypes, the clinical hold-out set will be supplemented with additional images drawn from open-access repositories such as the ISIC archives to meet the minimum threshold requirements. Because these supplementary images do not originate from Filipino respondents, the study will track and report the proportion of the hold-out set that is genuinely locally sourced versus supplementary, and will scope any claims regarding authentic Philippine clinical domain-shift robustness accordingly. To the extent that the hold-out set consists of genuine Filipino respondent submissions, it serves as the ultimate stress test for the proposed framework — aiming to evaluate whether a baseline

model's accuracy degrades when moving from standardized public repository data to the local Philippine subset, the ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework will evaluate whether the framework successfully bridges that performance gap.

The study focuses exclusively on images depicting one of the eight target infectious skin diseases included in the research. Images from both sources are eligible for inclusion if they carry a confirmed disease label corresponding to one of the eight target categories, fall within Fitzpatrick Skin Types I–V, and meet the study's quality standards. Facial images are excluded from the dataset regardless of source. Images that exhibit poor quality, excessive blur, extreme viewing angles, significant visual obstruction, or other conditions that may affect annotation and model training will likewise be excluded from the dataset.

The target dataset is expected to contain 300–500 images per disease subtype, resulting in an estimated total of 2,400–4,000 images across the eight selected skin diseases. To preserve dataset quality and representation, a minimum threshold of 300 verified images per disease subtype will be maintained. Disease subtypes that fail to satisfy this hard minimum threshold despite exhaustive searching across all available sources will be excluded from the study; given the deliberately clustered design, such an exclusion would require the research team to identify a substitute disease exhibiting comparable morphology within the same cluster in order to preserve the study's fine-grained classification objective. The final dataset will encompass fungal, viral, and bacterial skin disease presentations commonly encountered in Philippine dermatological practice and will serve as the primary resource for training, validating, and evaluating the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework.

Sampling Data

This study employs a purposive sampling technique integrated with stratified data splitting to construct the primary dataset for computational evaluation. Given the algorithmic objective to address Fine-Grained Visual Categorization (FGVC) challenges and morphological overlap across diverse skin tones, with a primary emphasis on melanin-rich skin (Fitzpatrick III–V), random sampling methodologies are statistically unviable. Consequently, all image data must be deliberately curated to satisfy strict clinical and visual criteria prior to being processed by the Adaptive ITA-CLAHE pre-processing module. To strictly comply with the Data Privacy Act of 2012 (RA 10173), the sampling process is delimited to fully anonymized data; every sampled image must be stripped of Personally Identifiable Information (PII). To be considered clinically valid, public repository images must possess verified source labels, while the locally sourced Clinical Hold-Out images must be strictly accompanied by a redacted, doctor-confirmed diagnosis.

Sample Size and Class Stratification

To specifically evaluate the Fine-Grained Visual Categorization (FGVC) problem and morphological overlap, the scope of the dataset is delimited to eight target infectious classes divided into two clusters: the vesiculopapular/eruptive cluster, Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, and Impetigo (*Impetigo Contagiosa (Mamaso)*); and the papulosquamous/verrucous cluster, Ringworm (*Tinea Corporis (Buni)*), An-an (*Tinea Versicolor*), Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*). The study establishes an optimal dataset threshold of 300 to 500 images per target class, projecting an aggregated dataset volume of 2,400 to 4,000 images. During the sampling phase, the dataset will be mathematically profiled to extract the baseline Individual Typology Angle (ITA) of each

image. This stratification ensures a sufficient variance of skin tones—from the highest ITA (lightest skin) to the lowest ITA (darkest skin)—which is a strict prerequisite for calibrating the $clip_limit=f(ITA)$ mathematical function. To evaluate the domain shift phenomenon, the dataset includes Fitzpatrick Skin Types I through V, which are categorically grouped into two distinct cohorts for comparative analysis: Group A (Fitzpatrick I–II) and Group B (Fitzpatrick III–V).

Stratified Data Splitting

To evaluate the specified hypotheses quantitatively and prevent data leakage, the curated dataset will be structurally divided into two independent branches: the Global Foundation (Public Dataset) and the Clinical Hold-Out Set.

The Global Foundation (Public Dataset), comprising 90–95% of the total image volume, will undergo a strict 70/20/10 stratified split across the eight target classes and Fitzpatrick cohorts:

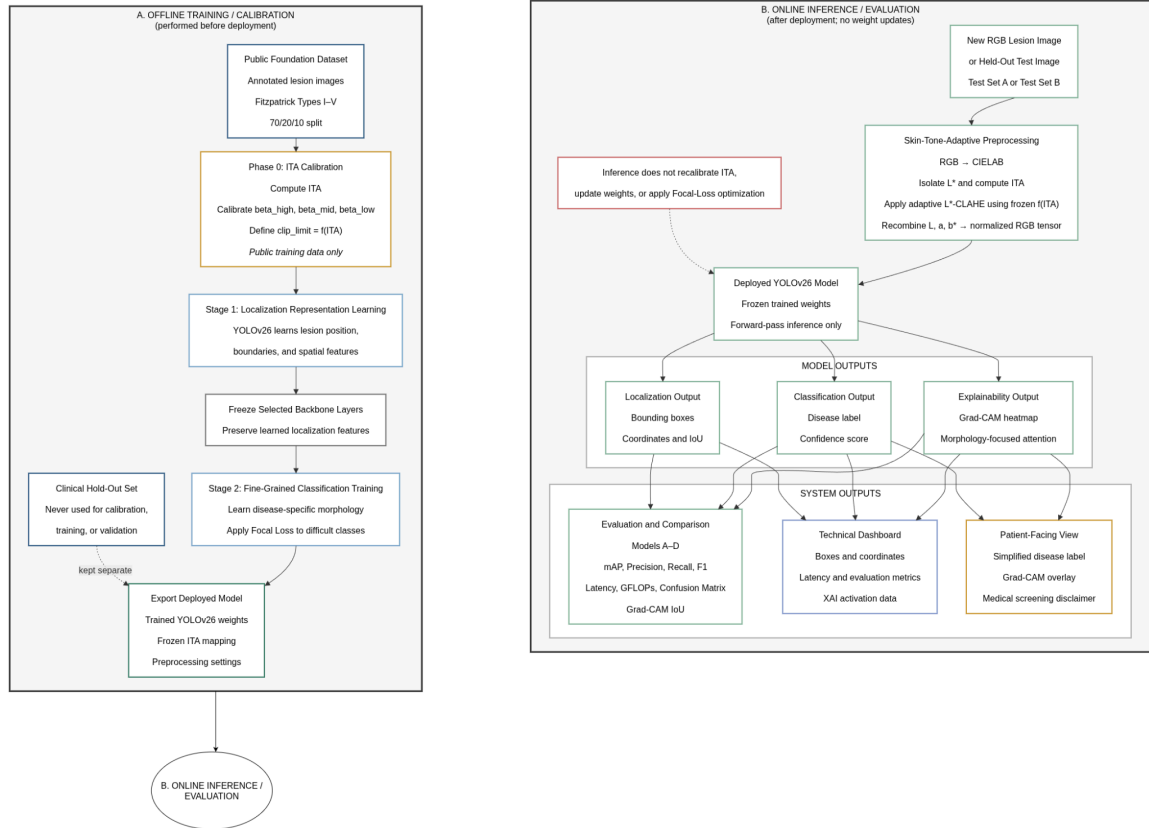
- **Training Set (70%):** Utilized exclusively for backpropagation, updating the network weights of the YOLOv26 architecture, and calculating the Focal Loss penalty.
- **Validation Set (20%):** Utilized during the training phase to monitor algorithmic overfitting and facilitate hyperparameter tuning.
- **Testing Set A (10%):** Kept strictly isolated from the training environment; utilized solely to calculate the baseline mean Average Precision (mAP), Precision, Recall, and Grad-CAM metrics on standardized public data.

The Clinical Hold-Out Set (Testing Set B), comprising 5–10% of the total dataset, is kept entirely outside the 70/20/10 split. It is never used during training or validation. It

functions exclusively as an external, out-of-distribution testing set to validate the model's robustness against real-world Philippine domain shifts.

System Architecture

Figure 8. System Architecture of the Study



Phase A encompasses all operations performed prior to deployment. The process begins with the Public Foundation Dataset, which consists of annotated lesion images across Fitzpatrick Skin Types I–V, rigorously partitioned into a 70/20/10 split. The initial computational step is Phase 0: ITA Calibration, where the Individual Typology Angle (ITA) is computed to empirically calibrate the enhancement thresholds (β_{high} , β_{mid} , and β_{low}). This phase exclusively utilizes the public training data to define the continuous mapping function $clip_limit=f(ITA)$. The framework then initiates a temporally decoupled

two-stage training strategy using the YOLOv26 architecture. Stage 1 is dedicated to Localization Representation Learning, where classification head is frozen, and only the bounding box regression losses (CIoU and DFL) and objectness loss are active so the network learns foundational spatial boundaries. To prevent feature degradation, the YOLOv26 CSP-Darknet backbone and localization heads are subsequently frozen. Stage 2 then proceeds with Fine-Grained Classification Training, unfreezing the classification head and allowing the model to focus entirely on disease-specific morphology while employing Focal Loss to penalize errors on difficult, minority classes. Throughout this entire offline phase, the Clinical Hold-Out Set remains strictly isolated and is never used for calibration, training, or validation. The phase concludes by exporting the deployed model, which packages the trained YOLOv26 weights, the frozen ITA mapping function, and all predefined preprocessing settings.

Phase B represents the deployed system utilized for online inference and evaluation, operating strictly without weight updates. When a new RGB lesion image or a held-out test image (from Test Set A or Test Set B (Clinical Hold-out Set)) enters the pipeline, it first undergoes Skin-Tone-Adaptive Preprocessing. The input tensor is mapped to the CIELAB color space to isolate the L^* channel. The system dynamically applies the identical k-means clustering (evaluating an empirical range of k values, by dynamically selecting the optimal k per image by maximizing the Silhouette Score, and explicitly extracting the cluster with the largest pixel area as the dominant healthy skin mask (with an automated fallback to a center-weighted bounding box crop if the extracted cluster's average luminance falls outside biologically plausible skin thresholds, followed by an automatic exclusion flag if the fallback also fails to detect plausible skin regions)). If flagged, the system halts processing, prompts the user via the

Patient-Facing View with a 'No plausible skin detected; please retake image' error, and explicitly bypasses the ITA calculation. Otherwise, it proceeds to extract the dominant healthy skin mask, and calculate its specific ITA value using the $ITA = \arctan\left(\frac{L^* - 50}{b^*}\right) \times \left(\frac{180}{\pi}\right)$. The system then applies adaptive L*-CLAHE enhancement utilizing the frozen clip_limit = f(ITA) logic, recombining the enhanced lightness channel with the preserved a^* and b^* chromatic channels to form a normalized RGB tensor. This tensor is fed into the deployed YOLOv26 model, which utilizes its frozen trained weights for forward-pass inference only; the system explicitly does not recalibrate the ITA thresholds, update model weights, or apply Focal Loss optimization during this phase. The inference engine generates three parallel Model Outputs: Localization Outputs (containing bounding boxes, coordinates, and IoU), Classification Outputs (providing the disease label and confidence score), and Explainability Outputs (generating Grad-CAM heatmaps that validate morphology-focused attention). Finally, these payload metrics are aggregated and routed into dedicated System Outputs. These outputs consist of an Evaluation and Comparison module that computes metrics such as mAP, Precision, Recall, F1-Score, inference latency, GFLOPs, confusion matrices, and Grad-CAM IoU across the ablated Models A–D. Simultaneously, the results are formatted for two distinct interfaces: a Technical Dashboard intended for clinical audit—displaying exact bounding box coordinates, latency metrics, and raw XAI activation data—and a Patient-Facing View that provides a simplified disease label, a Grad-CAM heatmap overlay, and a mandatory medical screening disclaimer.

Research Instrument

The primary research instrument used in this study is the *proposed ITA-Guided Adaptive L*-CLAHE and YOLOv26 Transfer Learning Framework*, developed to localize

and classify infectious skin lesions across Fitzpatrick Skin Types I–V, with emphasis on Types III–V. Implemented in Python, the framework integrates *CIELAB conversion*, *ITA calculation*, *adaptive L*-CLAHE*, *YOLOv26 object detection*, *transfer learning*, *sequential training*, *fine-grained morphological representation learning*, *Focal Loss*, and *Grad-CAM* into a unified pipeline for preprocessing, localization, classification, evaluation, and explainability.

The framework processes images through ITA calculation and profiling, empirical calibration of the ITA-to-CLAHE relationship, adaptive L*-CLAHE enhancement, YOLOv26 localization learning, backbone freezing, fine-grained morphological representation learning, Focal Loss classification, and Grad-CAM explanation. The adaptive enhancement follows $\text{clip_limit} = f(\text{ITA})$, while fine-grained learning focuses on texture, crusting, scaling, edges, shape, and lesion boundaries to improve discrimination among visually similar diseases.

A secondary research instrument is AnyLabeling, which is used for dataset preparation and annotation. It enables the creation and verification of bounding-box annotations and disease labels used as ground truth for model training, validation, and testing. Python-based scripts are also used for ITA computation, image preprocessing, adaptive CLAHE, dataset profiling, and automated evaluation.

The framework is evaluated using Precision, Recall, F1-Score, mAP₅₀, mAP_{50–95}, confidence scores, minority-class performance, inference latency, FPS, and GFLOPs. An ablation study compares four distinct configurations: Model A (Baseline YOLOv26), Model B (Fixed L*-CLAHE), Model C (Focal Loss), and Model D (the fully integrated Adaptive ITA-CLAHE proposed framework) to quantitatively evaluate the integrated and combined effects of the framework components.

Grad-CAM generates visual heatmaps to determine whether the model focuses on relevant lesion regions rather than background or illumination artifacts. System-generated outputs, including ITA values, calibrated clip limits, predictions, confidence scores, performance metrics, and Grad-CAM visualizations, provide quantitative and visual evidence for evaluating the framework's effectiveness in addressing illumination-induced feature degradation, morphological overlap, and long-tail class imbalance.

Lastly, for mobile deployment, the application will be designed to operate on smartphones running Android 10 or higher, with a 64-bit ARM-based processor, at least four CPU cores, 4 GB of RAM, 64 GB of internal storage with at least 4 GB available, an 8-megapixel rear autofocus camera, and a minimum display resolution of 720p. For the maximum or reference configuration, testing may be conducted on a smartphone running up to Android 17, with up to 12 GB of RAM, a 64-bit octa-core flagship processor such as the Snapdragon 8 Gen 2 or equivalent, 256 GB of internal storage, a 12-megapixel or higher rear autofocus camera, and a 1080p or higher display. The minimum device will be used to determine basic compatibility, while the maximum/reference device will be used to measure the best expected inference latency, Frames Per Second (FPS), memory consumption, battery usage, and device temperature. All final hardware specifications must be recorded from the actual smartphones used during testing.

Data Generation/Gathering Procedure

The skin lesion images used in this study will be sourced through a hybrid data acquisition strategy combining systematic curation from publicly available dermatological repositories and structured collection from locally sourced Philippine clinical

respondents, restricted throughout to the eight target diseases and Fitzpatrick Skin Types I-V (via image-based proxy). To ensure dataset quality, representation integrity, and methodological consistency throughout the research process, the dataset preparation will follow a structured curation, verification, calibration, and validation procedure.

1. Dataset Curation and Repository Acquisition. Images will be extracted and organized by disease category corresponding to the eight target infectious skin disease subtypes defined in the scope of this study: Molluscum Contagiosum, Chickenpox (*Varicella (Bulutong-tubig)*), Hand, Foot, and Mouth Disease, Impetigo (*Impetigo Contagiosa (Mamaso)*), Ringworm (*Tinea Corporis (Buni)*), *An-an (Tinea Versicolor)*, Warts (*Verruca (Kulugo)*), and Athlete's Foot (*Tinea Pedis (Alipunga)*). Only images with an existing disease label matching one of the eight target categories will be retained for further processing. All downloaded images will be catalogued with their source repository, original filename, disease label, and available metadata, including Fitzpatrick skin type classification where provided by the source dataset.

2. Clinical Hold-Out Set Collection. To constitute the clinical hold-out set — targeted at approximately 5–10% of the total dataset, an estimated 120 to 400 images distributed across the eight infectious skin disease subtypes, approximately 15 to 50 images per disease — data will be collected from Filipino respondents through a structured Google Form instrument. Respondents must be at least 18 years of age, have a doctor-confirmed diagnosis of one of the eight target infectious skin diseases, and may belong to any Fitzpatrick Skin Type I–V, with recruitment prioritized toward Fitzpatrick Skin Types III–V. Through the Google Form, respondents will provide their confirmed diagnosis and upload a photograph of their specific skin disease. All submitted images will be subjected to the same rigorous quality review applied to the public repository

datasets, ensuring that variations in mobile camera quality do not introduce unmanageable artifact noise into the out-of-distribution evaluation. The Google Form will include an explicit data privacy clause aligned with the provisions of the Philippine Data Privacy Act of 2012 (RA 10173), and participation will be entirely voluntary. The questionnaire instrument will be validated by at least three Registered Psychometricians (RPMs) prior to deployment. Ethics clearance from PUP Sta. Mesa will be obtained before any respondent data collection takes place.

3. Image Based Skin-Tone Grouping. Following initial acquisition from both sources, each image will undergo image-based proxy Fitzpatrick skin type verification and labeling. For images sourced from repositories that provide Fitzpatrick skin type metadata— skin type labels will be extracted directly from the accompanying metadata files and cross-referenced against the image records. For repositories where Fitzpatrick metadata is absent or incomplete, and for respondent-submitted images, the researchers will conduct manual visual verification, wherein each image is reviewed and assigned an image-based skin-tone grouping as a proxy for Fitzpatrick classification by the research team, acknowledging the inherent limitations of camera lighting variations. To ensure labeling reliability, manual proxy classification will be performed independently by at least two researchers, with any disagreements resolved through consensus or a third tie-breaking reviewer. Images confirmed to fall within Fitzpatrick Skin Types I–V will be retained; images whose skin tone cannot be reliably determined will be excluded from the dataset. To support the study's domain-shift evaluation, the retained dataset will be categorically grouped into Group A (Fitzpatrick Types I–II) and Group B (Fitzpatrick Types III–V), with recruitment and curation efforts prioritized toward Group B to preserve the study's melanin-rich analytical focus.

4. Image Quality Review. All curated and collected images will undergo a systematic quality review prior to ITA profiling and annotation. Images that are blurry, poorly illuminated, improperly framed, excessively low in resolution, or visually obstructed will be discarded. Only images that clearly show the lesion, meet the study's minimum resolution standard, and provide sufficient surrounding skin context for localization will be retained.

5. ITA Profiling and Adaptive Clip-Limit Calibration. Every image that passes quality review will undergo color-space conversion from RGB to CIELAB to extract its (L^*) and b^* channel values. The Individual Typology Angle (ITA) will then be computed exclusively on the dominant healthy skin regions, extracted via k-means clustering (evaluating an empirical range of k values, by dynamically selecting the optimal k per image by maximizing the Silhouette Score, and explicitly extracting the cluster with the largest pixel area as the dominant healthy skin mask (a proposed heuristic that will undergo visual validation during implementation)) on the image pixels to mathematically separate the lesion from the dominant skin tone to prevent data leakage prior to YOLO detection. The calibrated k-means segmentation masks will be manually verified by the researchers during the Phase 0 calibration to ensure the healthy skin baseline is accurately isolated from the lesion; any heuristic failure (for example, extracting background artifacts or shadows instead of skin) will be manually corrected by adjusting the k value for that specific image or excluding the image if separation remains impossible. The ITA is then calculated using $ITA = \arctan\left(\frac{L^* - 50}{b^*}\right) \times \left(\frac{180}{\pi}\right)$ (Ly et al., 2020). To resolve morphological overlap dynamically, the study utilizes a piecewise thresholding approach to define the $clip_limit=f(ITA)$ logic gate. The system applies specific clip limits based on established dermatological ITA classifications for melanin-rich and lighter skin (Osto et al., 2022):

1. If $ITA < 28$ (Darkest Skin): $clip_limit = \beta_{high}$
2. If $28 \leq ITA \leq 41$ (Medium Skin): $clip_limit = \beta_{mid}$
3. If $ITA > 41$ (Lightest Skin): $clip_limit = \beta_{low}$

Because determining the optimal enhancement thresholds is a primary objective of this study, the exact numerical values for β_{high} , β_{mid} , and β_{low} will be empirically established during the dataset calibration phase (Phase 0). To prevent data leakage, calibration will be conducted strictly using the public training dataset; the clinical hold-out set will never be used during calibration. The researchers will identify the maximum allowable enhancement limit for each ITA bracket that successfully exposes lesion morphology (crusts, scaling) before sensor noise amplification occurs. The calibration will utilize a grid search over candidate clip limit values (e.g., 2.0 to 10.0), selecting the highest β value for each ITA bracket that maximizes spatial contrast without exceeding a predefined noise amplification limit (provisionally set to a maximum local contrast variance of 50, to be finalized during calibration).

6. Hard Threshold Verification and Disease Class Balancing. Following ITA profiling, the researchers will audit the verified image count for each of the eight disease subtypes across both data sources. Each subtype must meet the hard minimum threshold of 300 verified images within the global public dataset, with a required minimum representation of at least 100 images per skin-tone cohort (Group A and Group B), before it is eligible for inclusion in model training. The study will aim to reach the target range of 300–500 images per subtype wherever dataset availability permits. Any subtype that fails to satisfy the 300-image hard threshold despite exhaustive searching will require the research team to either extend collection efforts specifically for that disease or identify a substitute disease of comparable morphology within the same cluster (for example,

substituting another viral skin disease if one disease falls short). Any substitution will be explicitly documented prior to model training to maintain the validity of the predefined research questions.

7. Image Annotation and Dataset Preparation. After curation, collection, threshold verification, and ITA profiling, the images will be organized by disease category and prepared for machine learning analysis. The researchers will review each image and identify the location of the lesion by drawing bounding boxes around the affected skin area using the AnyLabeling annotation tool. Images containing multiple lesions will receive separate annotations for each lesion. To ensure annotation accuracy, the entirety of the clinical hold-out set will be reviewed by a dermatologist, alongside a sampled portion of the public repository images. Any disagreements will be discussed and corrected before inclusion in the final dataset. Prior to partitioning, the dataset will undergo rigorous duplicate and near-duplicate removal, enforcing source-level separation to guarantee that related images do not leak across sets.

The global dataset will then be divided into three groups:

- Training Set (70%)
- Validation Set (20%)
- Testing Set (10%)

The clinical hold-out set is kept entirely separate from this split and is never used during training or validation. At evaluation time, the testing phase will therefore be conducted through two independent subsets: Test Set A (Public Distribution), comprising the 10% test-allocated images drawn from the global foundation dataset, and Test Set B (Clinical Hold-Out Set), comprising locally sourced Philippine clinical images, supplemented by global repository data where necessary. During evaluation, performance will be reported separately for the local Philippine subset and the global

supplementary subset to ensure claims regarding authentic Philippine clinical domain shift remain strictly accurate.

8. Final Dataset Validation. Prior to model development, the final dataset will undergo a comprehensive review to verify image quality, image-based proxy classification accuracy across Types I–V, ITA calibration integrity, annotation accuracy, and disease labeling consistency across both public and clinical data sources. The resulting dataset will serve as the primary resource for training, validating, and evaluating the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework. Model training will be executed using standard YOLO hyperparameters, including an initial learning rate of 0.01, SGD optimizer, an initial batch size of 16 (scalable up to 32 depending on hardware limitations), an input resolution of 640x640, and a maximum of 100 epochs (allocated as 50 epochs for Stage 1 localization and 50 epochs for Stage 2 classification, with learning rate decay applied independently per stage, and the learning rate reset to its initial value at the start of Stage 2) with early stopping patience set to 20 epochs.

Ethical Considerations

This study will observe ethical principles in the acquisition, curation, storage, and use of skin lesion images drawn from both publicly available dermatological repositories and locally sourced Philippine clinical respondents, covering Fitzpatrick Skin Types I–V and the eight target disease subtypes, with primary analytical emphasis on Fitzpatrick Skin Types III–V. The study involves a dual ethical framework: one governing the responsible use of secondary public dataset images, and another governing the collection of primary data from human respondents through a structured Google Form instrument.

For the public repository component, all images are sourced from publicly available repositories (specifically DermNet and Kaggle) whose data were originally collected under their respective ethical frameworks, institutional review processes, and participant consent procedures. The researchers will strictly observe the terms of use, licensing agreements, and attribution requirements stipulated by each repository. Images will be used exclusively within the boundaries permitted by each dataset's access agreement, and no images will be redistributed, commercialized, or used for purposes beyond those defined in this study.

For the clinical hold-out component, the study involves direct engagement with human respondents and is therefore subject to formal ethical review. Ethics clearance will be obtained from the PUP Sta. Mesa Institutional Ethics Review Committee prior to any respondent data collection. The Google Form instrument will include an explicit informed consent section, written in both English and Filipino, clearly explaining the purpose of the study, how the submitted images and diagnosis information will be used, the data privacy protections in place, and the respondent's right to withdraw participation at any time without consequence. Participation will be entirely voluntary, and eligibility is restricted to respondents who are at least 18 years of age with a doctor-confirmed diagnosis of one of the eight target diseases, spanning the full Fitzpatrick Skin Type I–V range, with recruitment prioritized toward Fitzpatrick Skin Types III–V. This age restriction simplifies the consent process by requiring only the respondent's own informed consent, without the need for a separate parental or guardian consent procedure. To verify clinical validity without compromising respondent privacy, respondents will be instructed to redact or digitally black out all personal identifying information (names, hospital addresses, contact numbers) from their medical certificates prior to upload, ensuring only the official diagnosis and the physician's generic license

validation remain visible for verification. The Google Form will incorporate a data privacy clause aligned with the provisions of the Philippine Data Privacy Act of 2012 (RA 10173) to ensure that all respondent-submitted data is handled in accordance with applicable legal and ethical standards. To directly address concerns regarding respondent privacy, the study confirms that no direct personal identifiers (names, addresses) will be collected, though redacted sensitive health information (diagnoses and certificates) will be securely processed. The only data retained from each submission is the clinical image itself and its corresponding doctor-confirmed disease label. Furthermore, all collected images will be stripped of embedded metadata (including EXIF data such as device information, geolocation tags, and timestamps) immediately upon collection to ensure that data is strictly de-identified in full compliance with the Data Privacy Act of 2012 (RA 10173).

The study also recognizes the importance of fairness and inclusivity in artificial intelligence. Because many existing dermatological AI systems perform poorly on darker skin tones, the proposed framework focuses on melanin-rich Filipino skin to promote more equitable model development. Model performance will be evaluated across Fitzpatrick Skin Types I through V (via image-based proxy), stratified by group, to identify potential biases and confirm that the system meaningfully improves outcomes for melanin-rich skin (Types III–V) relative to lighter skin tones, ensuring that the system does not unfairly favor lighter-skinned populations. This commitment to algorithmic fairness is consistent with emerging ethical standards for AI systems deployed in healthcare contexts, wherein performance equity across demographic groups is recognized as a fundamental ethical requirement (Obermeyer et al., 2019).

To maintain research integrity, all curated and collected images will undergo careful screening, verification, and annotation procedures before being included in the

final dataset. To securely manage the temporary use of Google Forms, email collection will be disabled, and the associated cloud drive will serve strictly as a restricted-access staging area. Upon submission, all dataset files, annotation records, respondent-submitted images, and associated research materials will be immediately transferred to and stored exclusively on encrypted external hard drives, followed by permanent deletion from the cloud staging area, to prevent unauthorized access and potential data leaks. These encrypted drives will be kept in a secure, controlled location accessible only to authorized researchers directly involved in the study. Upon completion of the research, all downloaded images, curated datasets, respondent-submitted images, and associated files will be permanently deleted from all storage devices in accordance with responsible data stewardship practices.

Results will be reported honestly and transparently, including any limitations, errors, or cases where the model performs poorly. The proposed framework serves as an AI-assisted screening and lesion localization tool designed to support dermatological expertise and does not provide diagnoses, prescribe treatments, or replace professional medical judgment. All predictions must be verified by qualified dermatologists before clinical application. The researchers are committed to the responsible development of artificial intelligence for dermatological applications, ensuring that the system's outputs are used in a manner that upholds patient safety, diagnostic accuracy, and equitable healthcare access.

All data used during the study will be employed exclusively for academic and research purposes. Images and related information will not be sold, shared with unauthorized parties, or used for purposes beyond those defined within the scope of this research. The researchers acknowledge that the original participants whose images appear in the public repositories consented to the use of their data for research, and this

study honors that consent by adhering strictly to the ethical and legal frameworks under which those datasets were made publicly available. For respondent-submitted clinical images, the same commitment to data protection and participant welfare applies, with all submissions handled in full accordance with the informed consent and data privacy provisions established prior to data collection.

Data Analysis

The data generated from the baseline YOLOv26 model and the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework will be analyzed quantitatively to evaluate their localization accuracy, fine-grained classification performance, interpretability, computational efficiency, and robustness across the full Fitzpatrick Skin Type I–V range, with primary analytical emphasis on Types III–V, applied to the eight target diseases across their two morphological clusters. The analysis will be conducted using predictions obtained from the testing dataset and compared against researcher-annotated ground-truth labels, supplemented by dermatologist validation strictly for the clinical hold-out set. The testing partition is further divided into two independent evaluation subsets: Test Set A, comprising the test-allocated images drawn from the global foundation datasets to establish baseline, in-distribution performance, and Test Set B (Clinical Hold-out Set), comprising the entirety of the locally sourced Philippine clinical images, supplemented by global repository data where necessary, to serve as the out-of-distribution domain shift stress test, with specific domain-shift conclusions restricted strictly to the local Philippine subset.

To address the first research question, the localization and classification performance of the baseline YOLOv26 model will be established using the testing dataset. Model outputs will be summarized through confusion matrices and class-wise

performance reports across the eight target diseases. Performance will be evaluated using Mean Average Precision (mAP₅₀ and mAP_{50–95}), Precision, Recall, F1-Score, Confidence Score Distribution, inference latency, Frames Per Second (FPS), and Giga Floating Point Operations (GFLOPs). To establish a quantitative benchmark for success, the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework (Model D) must demonstrate a statistically significant performance improvement over the Baseline YOLOv26 (Model A). Specifically, success is defined as achieving a higher Mean Average Precision (mAP₅₀) overall, demonstrating a measurable reduction in within-cluster misclassification rates among morphologically similar diseases, and yielding a statistically significant performance delta ($p < 0.05$) between the melanin-rich cohort (Group B) and the lighter skin cohort (Group A) under the Wilcoxon-Signed Rank Test. Because the eight diseases were deliberately selected to form two morphologically overlapping clusters, particular attention will be paid to the baseline confusion matrix's within-cluster misclassification rate—the rate at which the baseline model confuses diseases within the vesiculopapular/eruptive cluster (e.g., Varicella predicted as Mamaso, or Molluscum Contagiosum predicted as HFMD) or within the papulosquamous/verrucous cluster (e.g., Tinea Corporis predicted as Tinea Versicolor). This within-cluster misclassification rate serves as the primary empirical evidence of the Fine-Grained Visual Categorization problem motivating this study. To examine the effects of skin-tone variation, the testing dataset will be stratified according to the image-based proxy for Fitzpatrick Skin Types I through V, and mAP₅₀, mAP_{50–95}, Precision, Recall, and F1-Score will be computed independently for each. This stratified baseline analysis will expose the performance delta between lighter skin tones (Types I–II) and melanin-rich skin (Types III–V), quantifying the magnitude of the luminance-driven accuracy gap that the proposed framework is designed to close. Additionally, the baseline model will be evaluated separately on Test Set B (the clinical hold-out set) to

establish the magnitude of potential performance degradation experienced by the unmodified model when transitioning from standardized public repository data to the genuinely local Philippine subset of real-world clinical images.

To address the second research question, the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26 Framework will be evaluated using the same testing dataset, dataset partitions, and evaluation metrics to ensure a fair comparison with the baseline model. The framework integrates ITA-Guided Adaptive L*-CLAHE luminance normalization, sequential Stage 1/Stage 2 decoupled training, Focal Loss optimization, and Grad-CAM validation into a unified computational framework for infectious skin lesion localization. Similar to the baseline evaluation, performance metrics and the within-cluster confusion matrix will be computed for the proposed framework, allowing direct comparison of whether the adaptive algorithm specifically reduces confusion within each morphological cluster (for example, whether Varicella-Impetigo or Tinea Corporis-Tinea Versicolor misclassification rates decrease relative to baseline). To evaluate the integrated and synergistic effect of the full pipeline, an ablation study will be conducted across four model configurations: the Baseline YOLOv26 (Model A), YOLOv26 with Fixed L*-CLAHE using a standard clip limit of 2.0 (Model B), YOLOv26 with Focal Loss (Model C), and the fully integrated proposed framework (Model D). To ensure controlled baseline comparisons, Models A, B, and C will be trained using a standard, single-stage YOLOv26 protocol, while Model D exclusively utilizes the decoupled Stage 1/Stage 2 training procedure. Per-class Average Precision (AP50) across all eight disease subtypes will be recorded for each configuration to construct the data matrix required for the Friedman Test, with post-hoc Wilcoxon Signed-Rank testing with Bonferroni correction applied if significant differences are detected. Furthermore, the proposed framework will be evaluated independently on Test Set B (Clinical Hold-out

Set) to determine whether the ITA-Guided Adaptive L*-CLAHE enhancement and ITA-guided adaptive preprocessing can maintain localization and classification performance on the local Philippine subset where the baseline model degrades. Learning curves illustrating training loss, validation loss, and mAP progression throughout the training process will also be analyzed to assess learning stability and convergence behavior.

To address the third research question, the localization and classification performance of the proposed framework will be directly compared with that of the baseline YOLOv26 model under identical testing conditions. Because the dataset spans the full image-based proxy Fitzpatrick I–V range, the domain-shift-mitigation hypothesis is evaluated as originally designed. The performance delta ($\Delta AP50$), defined as the signed difference between the proposed framework and the baseline [$\Delta AP50 = AP50(\text{Model D}) - AP50(\text{Model A})$], will be computed independently for each of the eight disease classes within two cohorts: Group A (Fitzpatrick Types I–II) and Group B (Fitzpatrick Types III–V, the melanin-rich cohort this study is optimized for). A Wilcoxon-Signed Rank Test will then be applied to compare the median performance delta between these two paired groups. A p-value below 0.05 with a higher median rank for Group B will constitute limited, exploratory statistical evidence that the proposed framework provides significantly greater performance improvement for melanin-rich skin compared to lighter skin types, providing initial empirical evidence that the pipeline measurably provides greater improvement for the melanin-rich cohort, while acknowledging that $n = 8$ disease-level pairs remains a statistical limitation preventing conclusive proof of total bias removal. The clinical hold-out results of the Baseline (Model A) and the Proposed Framework (Model D) will likewise be presented side by side. A significantly smaller performance gap in the proposed framework relative to the

baseline will provide statistical evidence to evaluate whether the proposed pipeline can improve robustness against real-world domain shifts between standardized public data and the local Philippine subset. The results will be presented using comparative tables, graphs, and confusion matrices to illustrate the differences in localization accuracy, classification performance, within-cluster misclassification rates, and computational efficiency between Model A and Model D.

To evaluate model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) will be applied to both the baseline YOLOv26 model and the proposed framework. Grad-CAM heatmaps will be extracted specifically from the final convolutional layer of the YOLOv26 backbone to capture high-level spatial morphology. The generated activation heatmaps will be examined to determine whether the models focus on clinically relevant lesion regions (such as vesicle borders, crusting, or scaling) rather than surrounding skin, illumination artifacts, or background features. To distinguish genuine lesion-driven attention from attention artifacts potentially introduced by the L*-CLAHE enhancement process itself, Grad-CAM heatmaps will be generated and compared across three preprocessing conditions applied to the same underlying images: no enhancement (raw input), fixed-parameter static CLAHE, and the proposed ITA-Guided Adaptive L*-CLAHE. Because the proposed framework is trained exclusively on adaptively enhanced data, testing it on raw and fixed-CLAHE images is conducted strictly as an exploratory comparison rather than a matched evaluation. Heatmaps across all three conditions will be binarized using a static intensity threshold established during the validation phase and applied uniformly across all models to ensure strict comparability. A heatmap pattern that shifts progressively closer to the ground-truth lesion boundary moving from raw input to fixed CLAHE to adaptive CLAHE will be interpreted as evidence that the adaptive enhancement genuinely improves feature

visibility; a heatmap that instead shifts toward high-contrast edges or enhancement-induced noise regions unrelated to the ground-truth bounding box will be flagged as an enhancement artifact rather than genuine lesion attention. Pixel-attribution overlap (measured as the Intersection over Union (IoU) percentage serving strictly as a box-level feature attribution proxy rather than exact clinical morphological validation) will be computed and compared across all three preprocessing conditions, and across the baseline and proposed framework, spanning Fitzpatrick Types I through V. Descriptive statistics including mean and standard deviation of IoU scores will be reported per skin type, with higher mean IoU percentages indicating superior mitigation of luminance-driven attribution error, particularly within the Fitzpatrick III–V cohort. A successful interpretability validation requires the proposed framework to yield a higher mean IoU than the baseline model, consistently centering on true morphological structures (e.g., crusts, scaling) that the baseline model failed to detect. Grad-CAM heatmaps generated from Test Set B (Clinical Hold-out Set) images will receive particular attention, as the local Philippine subset of these images represents the most challenging domain shift condition and provide the most meaningful evidence of whether the proposed framework attends to genuine disease-differentiating morphology within each cluster (e.g., the umbilicated papule distinguishing *Molluscum Contagiosum* from the honey-crusted presentation of *Impetigo*, or the interdigital fissured scaling distinguishing *Tinea Pedis* from the verrucous surface of Warts) rather than skin-tone or illumination artifacts.

The findings will be presented through comparative tables, graphs, confusion matrices, learning curves, Grad-CAM visualizations, and statistical summaries. Collectively, these analyses will provide quantitative evidence regarding the effectiveness of the proposed ITA-Guided Adaptive L*-CLAHE Fine-Grained YOLOv26

Framework in resolving fine-grained morphological overlap among visually similar infectious skin diseases, while enhancing localization accuracy, robustness, and interpretability within the melanin-rich Filipino population.

Statistical Treatment

To analyze the empirical results and test the hypotheses formulated in this study, both descriptive and inferential statistical techniques will be applied at a significance level of $\alpha = 0.05$.

1. Descriptive Statistics

Descriptive statistics will be utilized to summarize and characterize the performance metrics of the four evaluated model configurations: Model A (Baseline YOLOv26), Model B (YOLOv26 + Fixed L* - CLAHE), Model C (YOLOv26 + Focal Loss), and Model D (Full Proposed Framework).

- **Localization and Classification Metrics.** The Mean, Standard Deviation (SD), and Median values across the eight target disease classes will be calculated for localization accuracy (mAP₅₀, mAP_{50–95}) and classification metrics (Precision, Recall, F1-score).
- **Demographic Stratification.** Metric scores will be aggregated and reported separately for Fitzpatrick Skin Types I–II and Fitzpatrick Skin Types III–V to establish baseline performance across skin tone categories.
- **Computational Efficiency.** Inference Latency (milliseconds per image), Frames Per Second (FPS), GFLOPs, and GPU Memory Consumption (MB) will be presented as descriptive summaries to evaluate model efficiency.

2. Inferential Statistics: Inter-Model Architecture Comparison

To determine whether statistically significant performance differences exist among the four model configurations across the eight disease classes, non-parametric statistical hypothesis testing will be conducted on paired per-class evaluation metrics.

Friedman Test

Evaluation metrics (mAP₅₀, mAP_{50–95}, Precision, Recall, F1-score) measured across identical disease classes represent non-independent, repeated-measures samples across four model configurations ($k = 4$), the non-parametric Friedman Test will be applied. The test statistic Q is computed as:

$$Q = \frac{12}{Nk(k+1)} \sum_{j=1}^k R_j^2 - 3N(k+1)$$

where N

represents the number of disease classes ($N = 8$), k represents the number of evaluated models ($k = 4$), and R_j is the sum of ranks assigned to model j .

Post-Hoc Pairwise Comparisons

If the Friedman test yields a statistically significant result ($p < 0.05$), a post-hoc Wilcoxon Signed-Rank Test with Bonferroni Correction will be executed for all pairwise model comparisons (specifically evaluating Model D against Model A, Model B, and Model C). The adjusted significance threshold α_{adjusted} for pairwise comparisons will be set to:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m} = \frac{0.05}{3} \approx 0.0167$$

where $m = 3$ represents the total number of pairwise model comparisons involving the proposed framework.

3. Inferential Statistics: Cross-Demographic / Skin Tone Domain Shift Comparison

To rigorously verify whether the proposed framework effectively resolves skin-tone bias and mitigates domain shift on melanin-rich skin, evaluation will focus on the relative performance improvement (Performance Delta, ΔAP_{50}) rather than raw accuracy scores alone.

Calculation of Performance Delta

For each of the eight independent disease classes ($N = 8$), the class-specific Performance Delta (ΔAP_{50}) is defined as the net gain in bounding box precision and classification accuracy achieved by the proposed framework relative to the baseline model:

$$\Delta AP_{50} = AP_{50, \text{Model D}} - AP_{50, \text{Model A}}$$

This delta metric isolates the actual performance boost generated by the proposed optimizations within each demographic cohort, allowing for a direct statistical comparison of improvement magnitudes across skin tone groups. To ensure transparent fairness evaluation, the absolute AP_{50} scores for both Group A and Group B will be reported alongside this performance delta.

Wilcoxon Signed-Rank Test

The evaluations will compare the performance gain (ΔAP_{50}) between the

two demographic cohorts: Group A (Fitzpatrick Skin Types I–II, $n_1 = 8$) and Group B (Fitzpatrick Skin Types III–V, $n_2 = 8$). Because the comparisons are matched by the exact same disease categories, the non-parametric Wilcoxon Signed-Rank Test will be employed.

The test evaluates the paired differences ($d_i = \Delta AP50_{(Group\ B)} - \Delta AP50_{(Group\ A)}$) for each disease. The test statistic W is computed by ranking the absolute values of these non-zero differences and summing the ranks of the positive and negative differences:

$$W = \sum_{i=1}^{N_r} [sgn(d_i) \cdot R_i]$$

where N_r is the number of non-zero differences, $sgn(d_i)$ is the sign of the difference, and R_i is the assigned rank.

Decision Rule. A statistically significant result ($p\text{-value} < 0.05$) with a positive median difference indicates that the proposed framework delivers a significantly larger performance boost ($\Delta AP50$) on melanin-rich skin compared to lighter skin tones, providing empirical evidence that the pipeline measurably provides greater improvement for the melanin-rich cohort.

Effect Size Estimation

To quantify the magnitude of the demographic performance difference independent of sample size, the Rank-Biserial Correlation (r_{rb}) will be calculated

for all outcomes:

$$r_{rb} = 1 - \frac{2U}{n_1 n_2}$$

The resulting r_{rb} value serves as a standardized measure of effect size, reflecting the practical strength of the framework's domain-shift mitigation capabilities.

4. Qualitative and Quantitative Interpretability Evaluation (Descriptive Analysis)

To evaluate diagnostic transparency and visual alignment, the Pixel-Attribution Overlap (Intersection over Union, or IoU %) from Grad-CAM heatmaps will be analyzed.

Metrics Tracked: Mean IoU (%).

Analysis: Descriptive statistics (mean, standard deviation) will be used to compare the baseline model's visual targeting against the proposed framework's targeting across individual image-based proxy Fitzpatrick skin types. Higher mean IoU percentages indicate superior mitigation of skin-tone bias through more accurate clinical feature prioritization (e.g., attending to morphological features like scaling or crusting rather than illumination artifacts or background noise).

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APPENDICES

Appendix 1: Experiment Papers

Experiment Paper 1

Per-Class Localization and Classification Performance (Wilcoxon Signed-Rank Test Data)

Objectives: To observe and record the Average Precision (AP) scores for each of the 8 disease classes between the baseline and other model configurations. This dataset serves as the input for the Friedman Test (raw scores) and the post-hoc Wilcoxon Signed-Rank Tests (pairwise performance deltas).

Procedure:

1. The evaluation script will parse bounding box predictions on the test dataset and compute class-specific Average Precision AP₅₀ and AP₅₀₋₉₅ scores for Models A, B, C, and D across all 8 disease classes (Experiment Paper 1A and 1B).
2. Class-specific performance differences ΔAP_{50} and ΔAP_{50-95} will be calculated across the identical 8 disease classes for the three planned pairwise comparisons: Model A vs. Model D, Model B vs. Model D, and Model C vs. Model D (Experiment Paper 1C).
3. The raw scores will be structured into an 8 x 4 matrix for the omnibus Friedman Test, and the paired differences ΔAP will be evaluated using post-hoc Wilcoxon Signed-Rank Tests with a Bonferroni-adjusted significance threshold.

Experiment Paper 1A: Raw Per-Class AP_50 Scores

Purpose: Collects raw baseline and intermediate AP_50 scores for all 4 models.
Used directly for the Friedman Test.

Disease Class (1–8)	Baseline Model A: AP_50	Model B: AP_50	Model C: AP_50	Proposed Model D: AP_50
Buni (Ringworm)				
Mamaso (Impetigo)				
[Remaining 6 Classes]				

Experiment Paper 1B: Raw Per-Class AP_50-95 Scores

Purpose: Collects strict bounding box overlap scores (AP_50-95) across all 4 models.

Disease Class (1–8)	Baseline Model A: AP_50–95	Model B: AP_50–95	Model C: AP_50–95	Proposed Model D: AP_50–95
Buni (Ringworm)				
Mamaso (Impetigo)				

<i>[Remaining Classes]</i>	6				
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Experiment Paper 1C: Pairwise Performance Deltas (Δ AP)

- **Purpose:** Computes the paired differences for the **Post-Hoc Wilcoxon Signed-Rank Tests** (A vs D, B vs D, C vs D).

Disease Class (1–8)	Model A vs D Difference (Δ AP ₅₀)	Model B vs D Difference (Δ AP ₅₀)	Model C vs D Difference (Δ AP ₅₀)	Model A vs D Difference (Δ AP _{+50–95})	Model B vs D Difference (Δ AP _{50–95})	Model C vs D Difference (Δ AP _{50–95})
Buni (Ringworm)						
Mamaso (Impetigo)						
<i>[Remaining 6 Classes]</i>						

Experiment Paper 2

Quantitative Metric Evaluation of Pipeline Components (Macro & Stratification)

Objectives: To compile the macro-level metric performance of the models across the entire dataset and stratified skin groups to address the descriptive statistics required by Research Question 1.

Cohort Stratification	Performance Metric	Model A	Model B	Model C	Model D
Global Dataset (All Tones)	mAP_{50}				
	mAP_{50-95}				
	Precision				
	Recall				
	F_1 -Score				
	Inference Latency (ms)				
	GFLOPs				
Group A (Fitzpatrick I-II)	mAP_{50}				
	mAP_{50-95}				
	Precision				
	Recall				

	F_1 -Score				
Group B (Fitzpatrick III–V)	mAP_{50}				
	mAP_{50-95}				
	Precision				
	Recall				
	F_1 -Score				

Experiment Paper 3

Skin-Type Domain Shift Mitigation (Wilcoxon-Signed Rank Test Data) Objectives:

To record and compare the class-specific Performance Delta (ΔAP_{50}) of Fitzpatrick Types I–II against Fitzpatrick Types III–V across all 8 independent disease classes to construct the dataset for the Wilcoxon-Signed Rank Test.

Disease Class (1–8)	Group A (Fitzpatrick I–II): ΔAP_{50}	Group B (Fitzpatrick III–V): ΔAP_{50}
1. Buni (Ringworm)		
2. Mamaso (Impetigo)		
3. [Class Name]		
[Remaining Classes 4–8]		

Experiment Paper 4

Qualitative and Quantitative Interpretability Evaluation (Grad-CAM)

Objectives: To evaluate model interpretability by measuring the Pixel-Attribution Overlap (IoU %) between Grad-CAM heatmaps and clinical ground-truth bounding boxes across skin types.

Image No.	Skin Group type	Ground Truth Class	Baseline Grad-CAM Overlap (IoU %)	Proposed Grad-CAM Overlap (IoU %)
001	Group A	Buni		
002	Group B	Mamaso		
<i>[Remaining Samples]</i>				